

Valorant Champions 2024 Seoul

Comprehensive Data Analysis Report

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Domain: Esports Analytics (Competitive FPS – Valorant)

Data Source : vlr.gg

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1. Executive Summary

This project analyzes the Valorant Champions 2024 (VCT Seoul) tournament to uncover what truly drives winning at the highest level of professional play. Using match, player, agent, map, and economy data, the analysis moves beyond raw kill counts to identify the structural, strategic, and efficiency-based factors that separate top teams from the rest.

The findings show that success is built on consistent player efficiency (ADR, ACS, KAST) rather than isolated fragging, map-aware agent and veto strategies rather than blind meta adoption, and disciplined economy management, where pistol and semi-buy rounds act as major momentum swing points. Teams that adapt compositions to map context, maintain balanced player impact, and extract value from lower-investment rounds consistently outperform those reliant on star performances or high spending alone.

By combining strong data engineering, exploratory analysis, and domain-driven interpretation, this project demonstrates how competitive Valorant outcomes can be explained through repeatable, data-backed decision patterns—offering actionable insights for analysts, coaches, and teams seeking a strategic edge.

2. Project Objective & Business Context

2.1 Objective

The primary objective of this project was to build a structured, repeatable analytical framework capable of explaining why certain teams consistently win at the highest level of competitive Valorant. Instead of relying on highlight-driven narratives or raw statistics, the analysis was designed to uncover the underlying performance and strategic drivers of success across three core dimensions:

- **Player Performance & Efficiency:** Identifying which individual metrics (e.g., ADR, ACS, KAST, clutch efficiency) best represent true round impact and consistency, and how different roles trade fragging power for utility and survivability.
- **Agent Meta & Map Strategy:** Evaluating how agent effectiveness varies by map, how veto and decider strategies reflect risk management, and why situational compositions outperform rigid meta imitation.
- **Economy & Round Win Impact:** Measuring how spending patterns, pistol rounds, semi-buys, and thrifty wins influence momentum, map win probability, and long-term match outcomes.

By focusing on efficiency, adaptability, and decision quality, the project avoids overvaluing isolated events and instead explains success as a result of repeatable, data-backed behaviors.

2.2 Business Context

In professional esports, teams operate under constraints similar to traditional businesses: limited resources, high variance environments, and the need to make fast, high-impact decisions under pressure. In the competitive environment teams get very few chances to prove themselves. Coaches, analysts, and management must decide how to allocate practice time, build rosters, select strategies, understand team’s and player’s strengths and weaknesses and manage risk—often with incomplete or noisy information or based on intuition.

This analysis mirrors real-world esports decision-making use cases, including:

- Roster evaluation and player role assignment
- Agent composition planning and map veto preparation
- Economy strategy optimization and momentum management
- Identifying inefficiencies that can be exploited against opponents

By translating tournament data into clear, actionable insights, this project demonstrates how data analytics can support competitive decision-making, reduce reliance on intuition alone, and provide measurable strategic advantages in high-stakes esports environments.

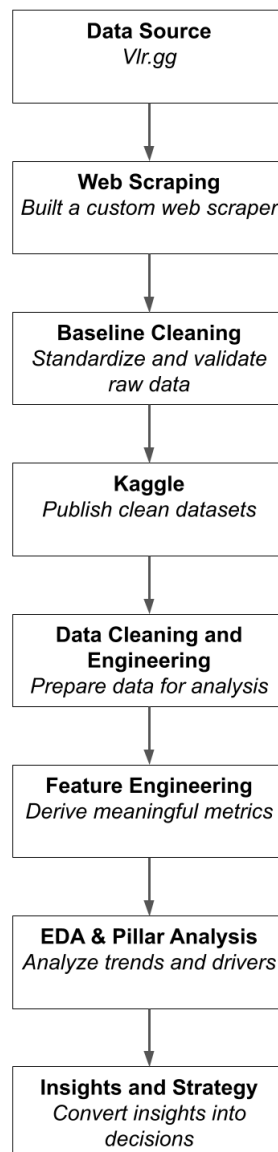
2.3 About Tournament

The 2024 Valorant Champions was the fourth annual world championship tournament for Valorant, a competitive first-person shooter game developed by Riot Games. The event brought together the 16 best teams from around the world—four each from the Americas, EMEA, Pacific, and China—who had qualified through rigorous regional and international competition throughout the season. The tournament began with a group stage where teams were divided into four groups of four and competed in a double-elimination format, meaning each team had to lose twice before being eliminated. The top two teams from each group advanced to the playoffs, which also followed a double-elimination bracket, with most matches played as best-of-three series and the lower bracket final and grand final played as best-of-five. The tournament concluded with Edward Gaming defeating Team Heretics 3–2 in a thrilling grand final—earning China’s first international Valorant championship and an MVP performance from Zheng “ZmjjKK” Yongkang.

Place	Team	Place	Team
1st Place	EDward Gaming	9th–12th Place	KRÜ Esports
2nd Place	Team Heretics	9th–12th Place	Gen.G Esports
3rd Place	Leviatán	9th–12th Place	Team Vitality

4th Place	Sentinels	9th–12th Place	Paper Rex
5th–6th Place	DRX	13th–16th Place	Bilibili Gaming
5th–6th Place	Fnatic	13th–16th Place	FunPlus Phoenix
7th–8th Place	Trace Esports	13th–16th Place	TALON Esports
7th–8th Place	G2 Esports	13th–16th Place	FUT Esports

3. Analytical Flow Diagram



4. Data Sources & Project Structure

4.1 Data Sources

The dataset is composed of multiple structured CSV files, each representing a specific analytical layer of the Valorant Champions 2024 (VCT Seoul) tournament. This modular design enables efficient joins and supports analysis at multiple granularities, ranging from match-level outcomes to player performance, agent usage, and in-game economy dynamics.

Core Reference & Metadata Files

- **event_info.csv**
Contains general information about the tournament, including event name, location, and format. This file serves as the top-level metadata reference for the analysis.
- **columns_description.csv**
Provides standardized definitions for all columns across the dataset. This file improves interpretability, ensures consistency across analytical workflows, and supports long-term maintainability of the dataset.

Match & Map-Level Data

- **matches.csv**
Stores match-level information such as match identifiers, participating teams, and final outcomes. This dataset functions as the primary linkage layer for integrating player, map, and economy-related data.
- **detailed_matches_maps.csv**
Contains map-specific information for each match, including map names, results, and team-level performance metrics. This dataset enables map-wise performance evaluation and analysis of terrain-specific trends..
- **maps_stats.csv**
Provides aggregated statistics at the map level, including attack and defense win rates. This file supports strategic analysis of map balance and side-specific advantages.

Player-Level Performance Data

- **player_stats.csv**
Provides overall player statistics for the tournament, offering a high-level view of individual performance and enabling leaderboard creation and outlier detection.

- **detailed_matches_player_stats.csv**
Contains granular, match-level player metrics such as ACS, ADR, KAST%, and K/D ratios. This dataset is central to evaluating player efficiency, consistency, and impact across different maps and matches..
- **performance_data.csv**
Includes advanced performance indicators such as multi-kill rounds and clutch situations. This dataset supports deeper analysis of high-impact moments that are not fully captured by standard statistical measures.

Agent & Strategic Data

- **agents_stats.csv**
Contains agent-level statistics, including pick rates and win rates across maps. This dataset enables analysis of agent popularity, effectiveness, and agent–map synergies within the tournament meta.

Economy Data

- **economy_data.csv**
Captures team-level economy information at the map and round level, including credits spent and buy types. This dataset is critical for analyzing the relationship between economic decision-making and round win probability, including the impact of Thrifty and low-buy rounds

4.2 Data Collection Methodology

The data for this project was collected through a custom-built web scraping pipeline developed specifically to extract professional match statistics from vlr.gg, one of the most authoritative and widely referenced sources for Valorant esports data. The pipeline was designed to capture match, map, player, agent, and economy-level information in a structured and analysis-ready format

The scraping process was implemented as a standalone data engineering project, demonstrating full ownership of the data acquisition layer. The scraper outputs normalized CSV files optimized for downstream analytical workflows.

- **Scraper Repository:** https://github.com/Piyush86kumar/Valorant_vlr.gg_scrapper
- **Output Format:** Structured CSV files with consistent schema and identifiers

Following data collection, the dataset underwent initial validation and baseline preprocessing to correct structural inconsistencies, standardize formats, and ensure schema reliability. This

cleaned version of the dataset was then published to [Kaggle](#) to enable community access, reproducibility, and collaborative analysis.

In addition to baseline preprocessing, a second level of analytical cleaning and transformation was performed during feature engineering. This stage involved reshaping data, correcting logical inconsistencies, and deriving analytical features required for player performance, agent meta, and economy impact analysis.

4.3 Project Structure

The project is organized to clearly separate data storage, analysis notebooks, and documentation, ensuring modularity and ease of navigation.

VCT 2024/

— VCT_2024_seoul_dataset/	# Dataset storage
— agents_stats.csv	# Agent utilization and win rates
— columns_description.csv	# Metadata and column definitions
— detailed_matches_maps.csv	# Specific map data per match
— detailed_matches_overview_processed.csv	# Cleaned/Transformed match overview
— detailed_matches_overview.csv	# Raw match overview data
— detailed_matches_player_stats.csv	# Match-by-match player performance
— economy_data.csv	# Team economy and buy-type stats
— event_info.csv	# VCT tournament general info
— maps_stats.csv	# Map win % (Attack vs. Defense)
— matches.csv	# List of all matches played
— performance_data.csv	# Clutches and multi-kill data
— player_stats.csv	# Aggregated player statistics
— Agents_stats.ipynb	# Analysis of agents_stats.csv
— Detailed_matches_maps.ipynb	# Analysis of maps and match flow
— Detailed_player_stats.ipynb	# Analysis of performance_data.csv
— Economy_stats.ipynb	# Analysis of economy_data.csv
— Player_stats.ipynb	# Analysis of player_stats.csv
— README.md	# Project documentation
— VCT 2024 Data Analysis Report.pdf	# Project report
— requirements.txt	# Necessary Python libraries

5. Data Engineering & Preparation

5.1 Data Integration

The raw data available on vlr.gg is distributed across multiple pages and views, making direct analysis impractical. To address this, a modular data integration strategy was implemented in which match, map, player, agent, and economy data were captured and stored in separate, well-defined CSV files.

This modular design allows each dataset to be analyzed independently while still supporting seamless cross-referencing. Rather than consolidating all data into a single table, the structure prioritizes flexibility and scalability across different analytical objectives

A key principle of the integration layer was interoperability across datasets. Consistent identifiers such as `match_id` and `player_id` were preserved across tables, enabling reliable joins and selective integration across analytical domains.

Normalizing Nested and Unstructured Fields

Some fields were originally stored in nested or unstructured formats unsuitable for relational analysis. For example, map utilization data embedded within single fields was normalized into structured tabular formats and aligned with agent- and map-level datasets, enabling consistent joins and cross-analysis.

Linking Match-Level and Round-Level Data

Match-level datasets were designed to integrate cleanly with round-level data using shared identifiers. Preserving `match_id` allowed match records to be enriched with granular round information when required, supporting both high-level and detailed analysis without sacrificing modularity.

By maintaining consistent identifiers and a modular schema, the data integration framework ensures consistency, scalability, and reliability. This foundation supports flexible analysis across player, team, map, and economy dimensions while preserving data integrity.

5.2 Data Preprocessing and Transformation

Following data integration, the raw datasets were preprocessed and transformed into a structured, analysis-ready format. This stage focused on cleaning irregular string representations, handling missing values, parsing complex fields, and reshaping data into standardized tabular structures. These steps ensured consistency and accuracy across datasets and enabled reliable downstream analysis.

Parsing and Expanding Complex Fields

Some columns contained complex data embedded within single fields. For example, the `map_utilizations` column initially stored dictionary-like information as strings. This data was

expanded into structured columns and aligned with agent-level attributes, enabling clear analysis of map utilization patterns alongside agent features.

Removing Redundant Attributes

Non-essential columns such as date, patch, match_title, and event were removed, as they did not contribute to the analytical objectives. This step reduced noise and ensured datasets remained focused on match-, round-, and performance-level features.

Team and Score Restructuring

Fields containing multiple values were decomposed into separate columns. The teams column was split into individual team fields, and the score column was separated into numeric values for each team. This restructuring enabled direct team-level comparisons and reliable computation of match outcomes.

Standardizing Entity Names

Team names appearing in different formats (e.g., abbreviations and full names) were standardized to a single naming convention. This ensured consistency across datasets and prevented mismatches during merges and visualizations.

Cleaning and Correcting Economy Data

In the economy_data dataset, economy-related fields stored rounds played and won together in string formats (e.g., "3 (2)"). These were split into separate numeric columns to enable accurate win-rate calculations. Additionally, pistol rounds were explicitly separated from eco rounds to prevent misclassification and improve the accuracy of economy-based analysis.

Completing Team–Map Coverage

To address missing team–map combinations, a complete team–map grid was generated and missing values were filled with zeros. This ensured consistent coverage across all teams and maps, eliminating structural gaps that could bias comparisons.

Through these preprocessing and transformation steps, the raw data was standardized into a clean and consistent format. By normalizing complex fields, correcting classifications, and addressing missing structures, this stage established a robust foundation for feature engineering and analytical evaluation.

5.3 Feature Engineering

Feature engineering enriched the dataset with normalized metrics, composite indicators, and efficiency measures, enabling consistent player- and team-level comparisons across matches, maps, and buy categories.

Pick and Veto Normalization

Map pick and ban information was originally stored as unstructured text. These entries were normalized into structured fields, enabling systematic analysis of team pick and ban preferences across different match formats.

Map Ban Frequency Engineering

Map bans initially appeared across multiple columns for each team. These fields were reshaped into a unified structure to compute ban frequencies and normalized ban percentages per map, allowing clear comparison of commonly banned maps across teams.

Composite Player Performance Scoring

A composite player performance score was engineered using weighted metrics such as first kills, deaths, ADR, and APR. This score provided a holistic measure of player impact beyond single-stat evaluations and enabled consistent ranking across multiple performance dimensions.

Thrifty Round Efficiency Metrics

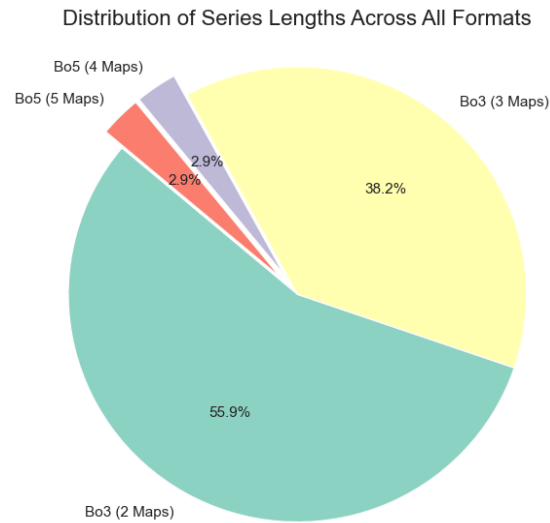
Thrifty rounds were defined as a combination of eco and semi-eco rounds. Team-level win rates were calculated for thrifty and full-buy rounds, with minimum participation thresholds applied to ensure statistical relevance. This enabled meaningful comparison of team efficiency under different economic conditions.

By normalizing raw fields, combining related indicators, and engineering efficiency-focused metrics, this stage transformed cleaned data into analysis-ready features. These engineered features form the foundation for exploratory analysis and pillar-based insights presented in subsequent sections.

6. Exploratory Data Analysis Overview

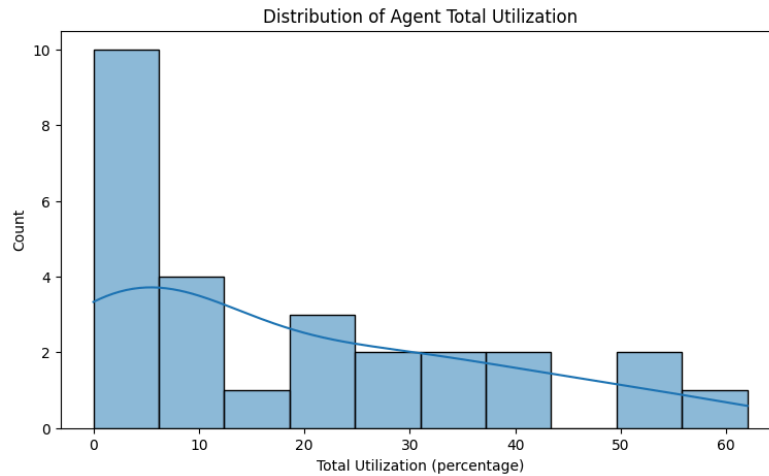
Exploratory Data Analysis was conducted to understand key patterns, distributions, and relationships within the dataset before applying deeper analytical techniques. This stage focused on player performance, team behavior, agent usage, map dynamics, and economy structure. By combining descriptive statistics with visual exploration, EDA helped validate expected trends while also surfacing hidden insights driven by tournament format and meta dynamics.

- **Distribution Of Series Length Across All Formats**



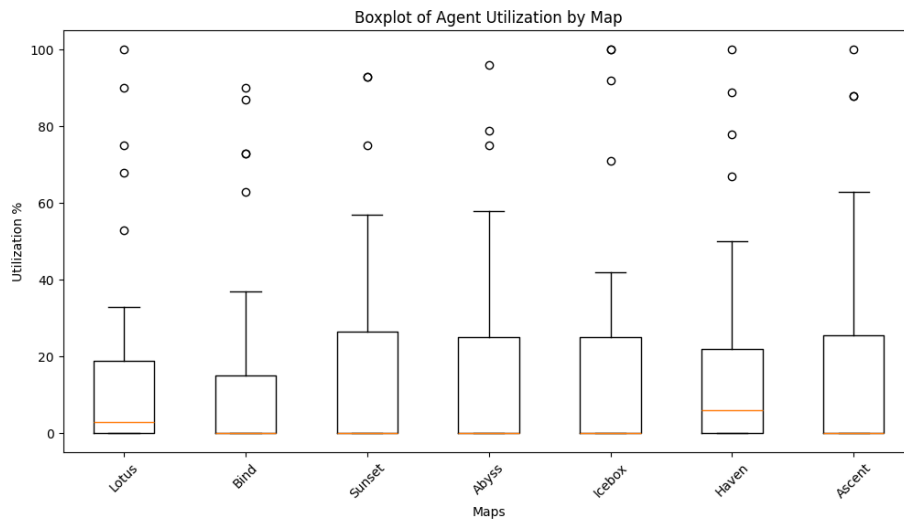
The distribution reflects the tournament structure, where the majority of matches were played as Best-of-3 (Bo3), with only the Lower Final and Grand Final conducted as Best-of-5 (Bo5). As a result, 2-map outcomes dominate the distribution, indicating frequent 2–0 victories—particularly in early rounds—where skill gaps between teams were more pronounced. The limited presence of longer series explains the small proportion of extended matches toward the latter stages of the tournament.

- **Distribution of Agent Total Utilization**



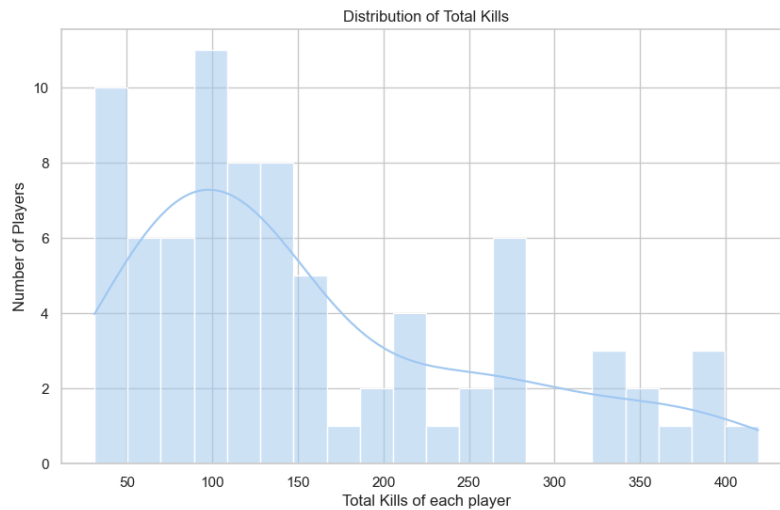
The agent utilization distribution is heavily right-skewed, with most agents falling in the 0–10% usage range. This indicates that a small subset of agents dominates the meta, while the majority are picked infrequently. The steep drop-off highlights limited strategic diversity and suggests that team compositions are strongly influenced by prevailing meta preferences rather than evenly distributed agent viability.

- Boxplot of Agent Utilization of Map**



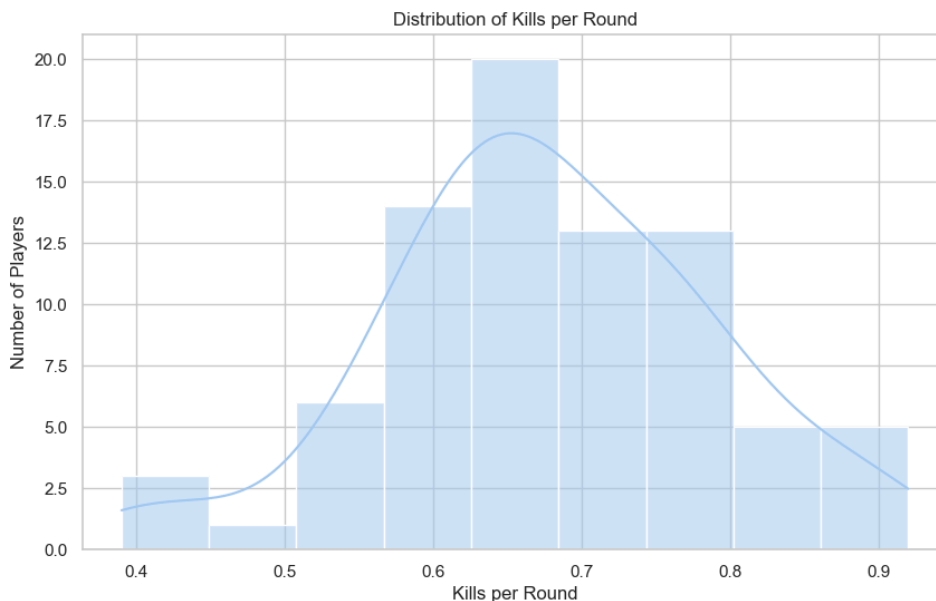
The boxplot of agent utilization by map reinforces the insight from the earlier histogram: most agents are underutilized overall, and their usage varies significantly depending on the map. While the histogram revealed a general skew toward low utilization, this chart doubles down by showing that some maps—like Sunset and Abyss—encourage broader agent diversity, whereas others—like Bind and Haven—favor more consistent picks. The circles are not outlier but represent the true nature of meta where a handful of agents are widely used on each map by most of the teams. Together, these visuals highlight how both overall meta and map-specific dynamics shape agent selection strategies.

- **Distribution of Kills**



The total kills distribution shows a clear right-skew, with most players clustered between 50 and 150 kills and a long tail extending toward higher values. This pattern is driven by tournament progression: players on teams eliminated early accumulated fewer kills due to limited matches, while players on deeper-running teams benefited from more rounds and maps. As a result, raw kill totals are strongly influenced by match volume rather than pure individual performance.

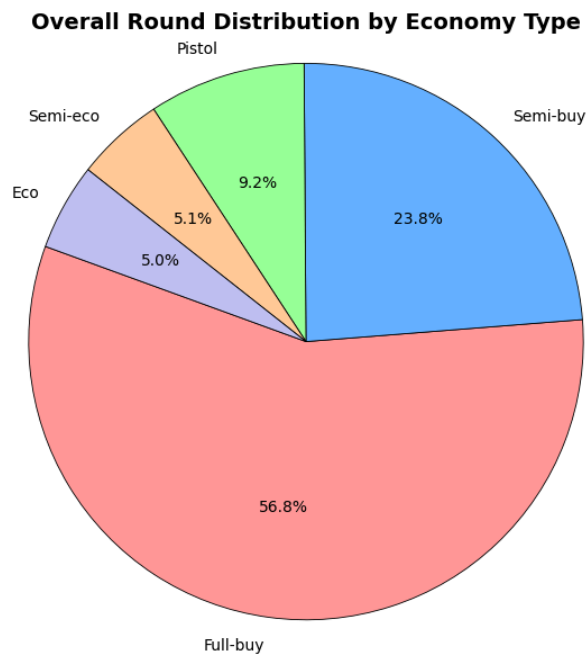
- **Distribution of Kills per Round**



Kills per round present a more normalized performance metric, with most players centered around ~0.65 kills per round. The slight right skew highlights a small group of consistently high-impact fraggers. This metric offers a fairer comparison across players by accounting for

differences in match length and reinforces the limitations of relying solely on aggregate kill counts.

- **Round Distribution by Economy Type**

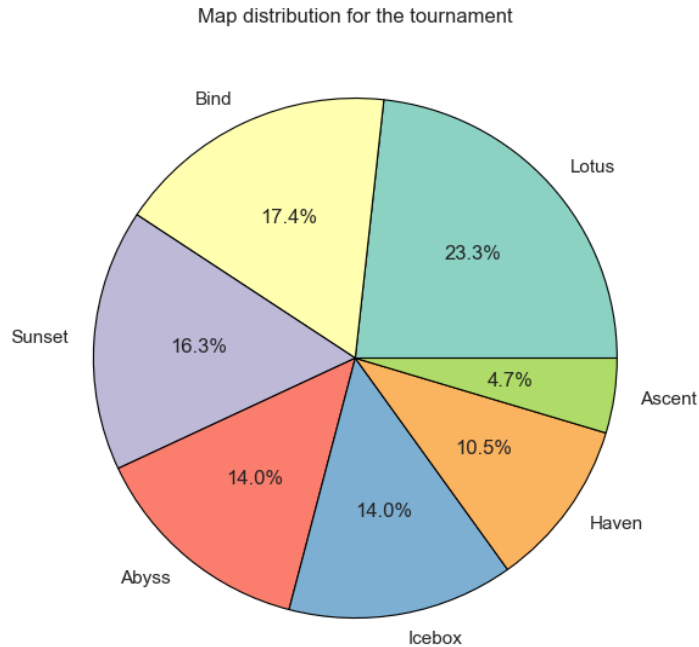


Economy Definations:

- Pistol: Early rounds with pistols only.
- Eco: No significant spending, default pistols only.
- Semi-Eco: Very light spending, saving for next round
- Semi-Buy: Partial investment, mixing cheaper weapons with some utility.
- Full-Buy: Strongest setup with rifles, armor, and all utility.

The distribution shows that Full-Buy rounds dominate (56.8%), underscoring their importance in competitive play. Teams consistently prioritize reaching full-buy states due to their higher round-win probability. Semi-Buy rounds (23.8%) reflect frequent transitional strategies to maintain competitiveness while stabilizing the economy. Pistol rounds (9.2%) are structurally fixed, while Eco and Semi-Eco rounds (~10%) occur less frequently and are primarily recovery mechanisms.

- **Map Distribution for the tournament**



Map selection was not evenly distributed across the tournament. Lotus (23.3%), Bind (17.4%), and Sunset (16.3%) collectively accounted for over half of all matches played. This concentration highlights their central role in team preparation and meta adaptation, while less-played maps likely reflect either strategic avoidance or lower comfort levels among teams..

Conclusion

Overall, the exploratory analysis highlights how tournament structure, meta preferences, and economic constraints shape player performance and team strategies. Key findings include the dominance of Bo3 formats, concentrated agent usage driven by meta rigidity, map-dependent strategic diversity, and the strong influence of match volume on raw performance metrics. These insights establish a strong foundation for deeper team-level, player-level, and economy-focused analyses in subsequent sections.

7. Pillar 1: Agent Meta & Map Strategy

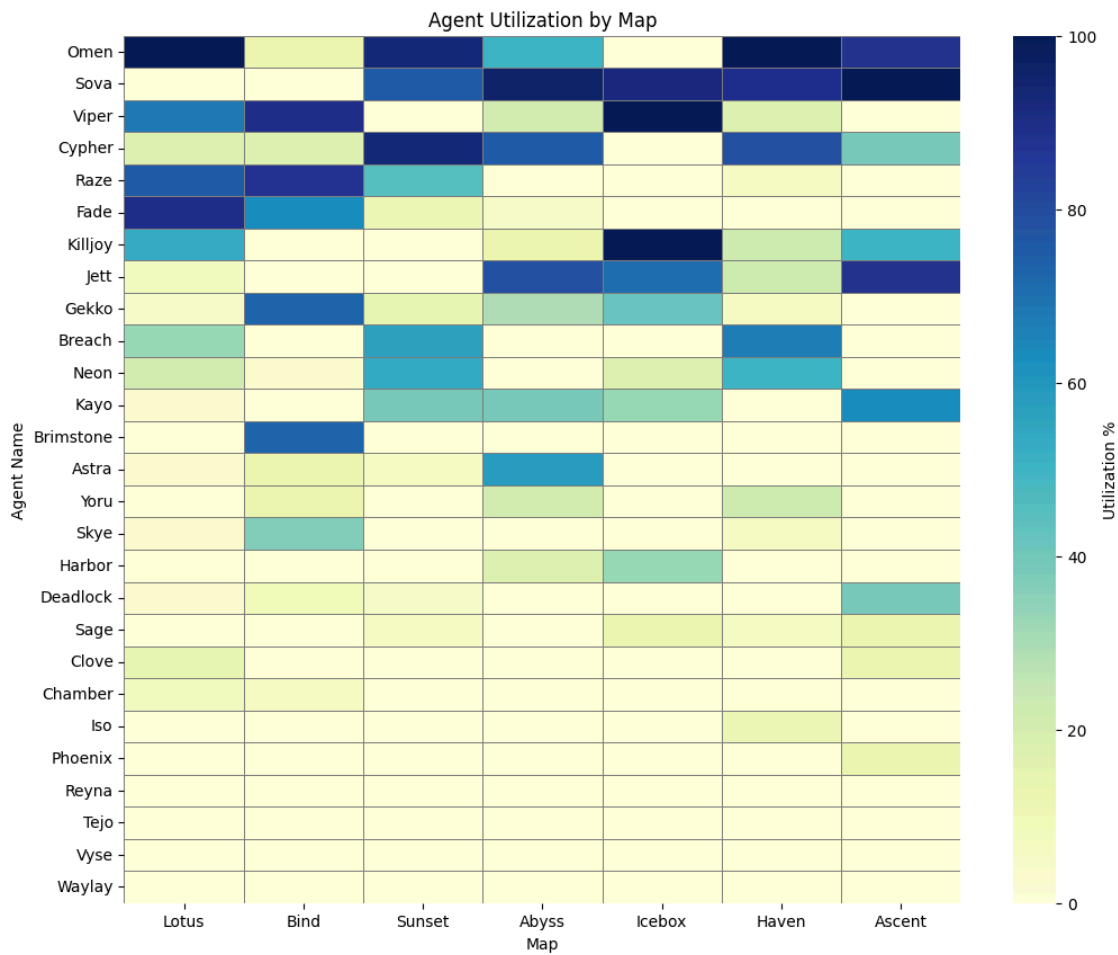
(*Agents_stats.ipynb, Detailed_matches_maps.ipynb*)

7.1 Analysis Focus

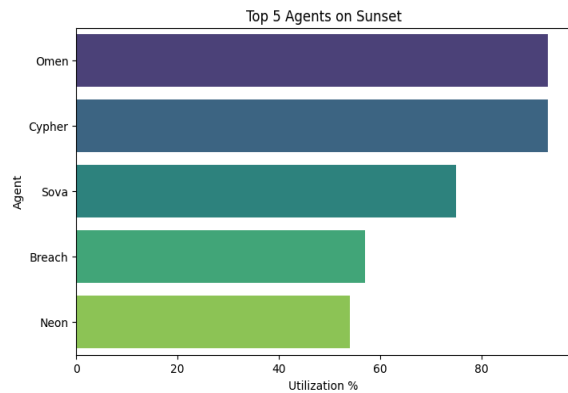
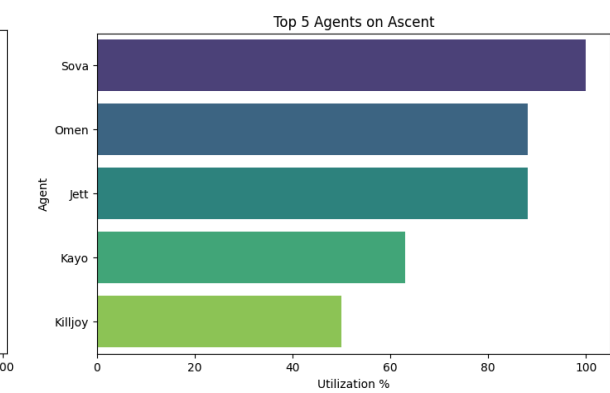
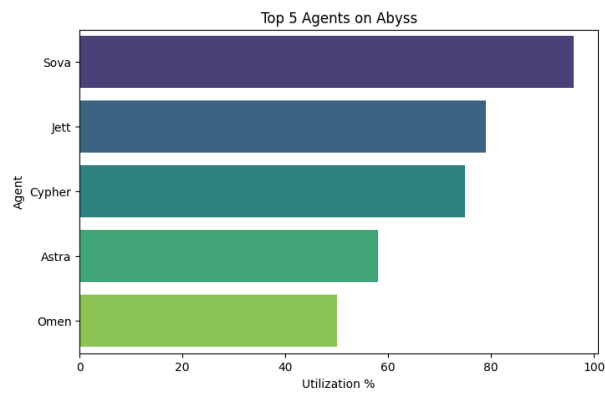
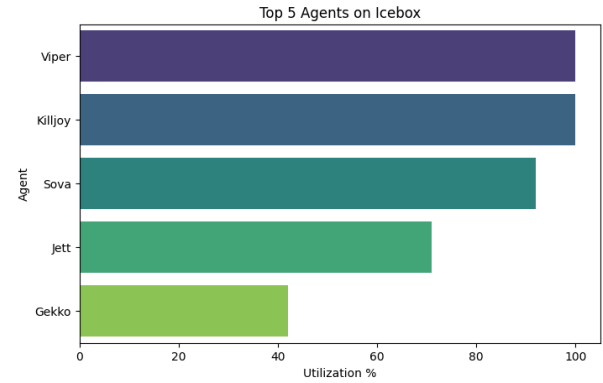
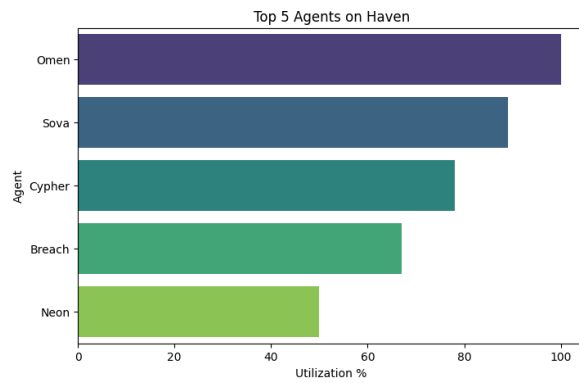
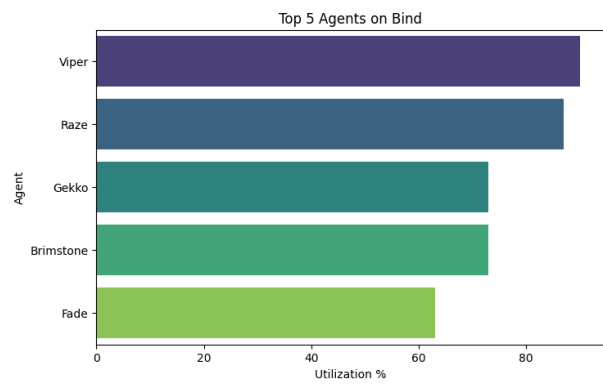
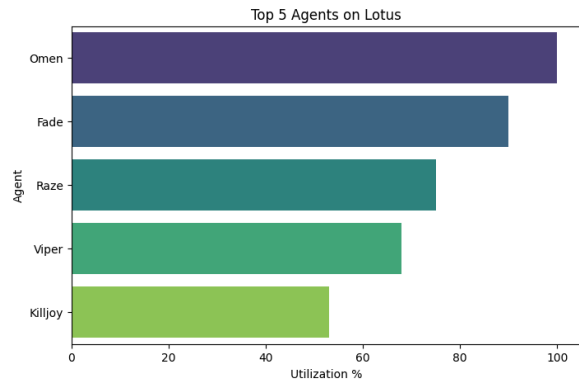
- Agent utilization by map
- Agent effectiveness (kills, headshots, assists, KAST, rating)
- Map pick/ban behavior and decider trends
- Composition diversity (Mirror vs Non-Mirror)
- Map pacing, side bias, and post-plant outcomes

7.2 Analysis

- **Agent Utilization by Map**



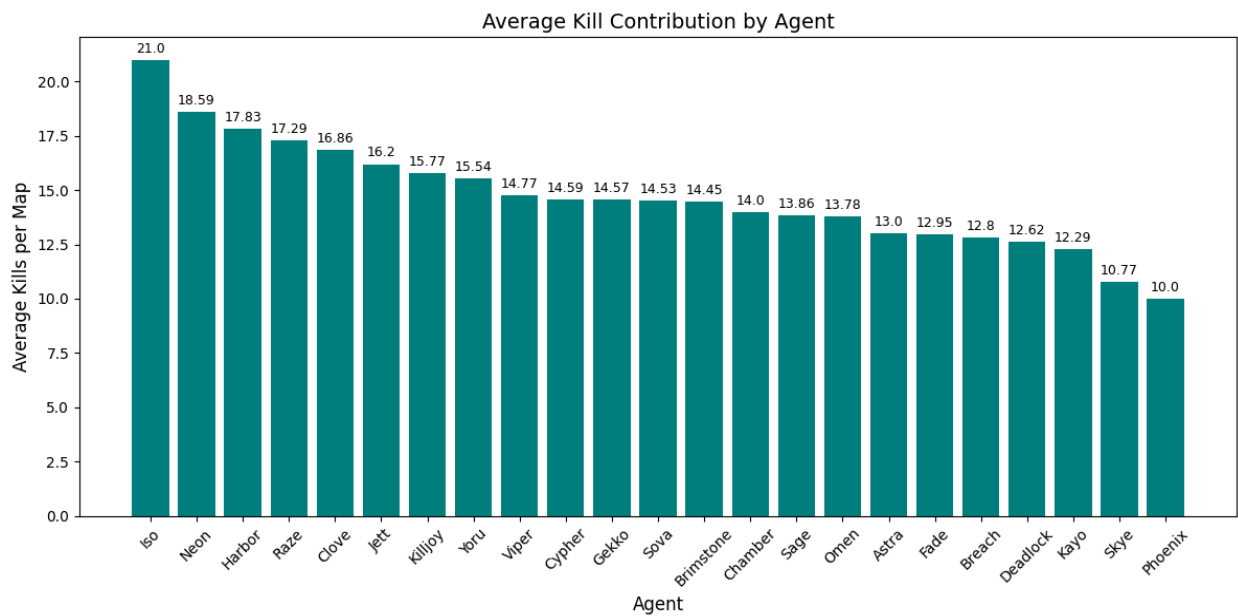
- **Top 5 Most Utilized Agents per Map**



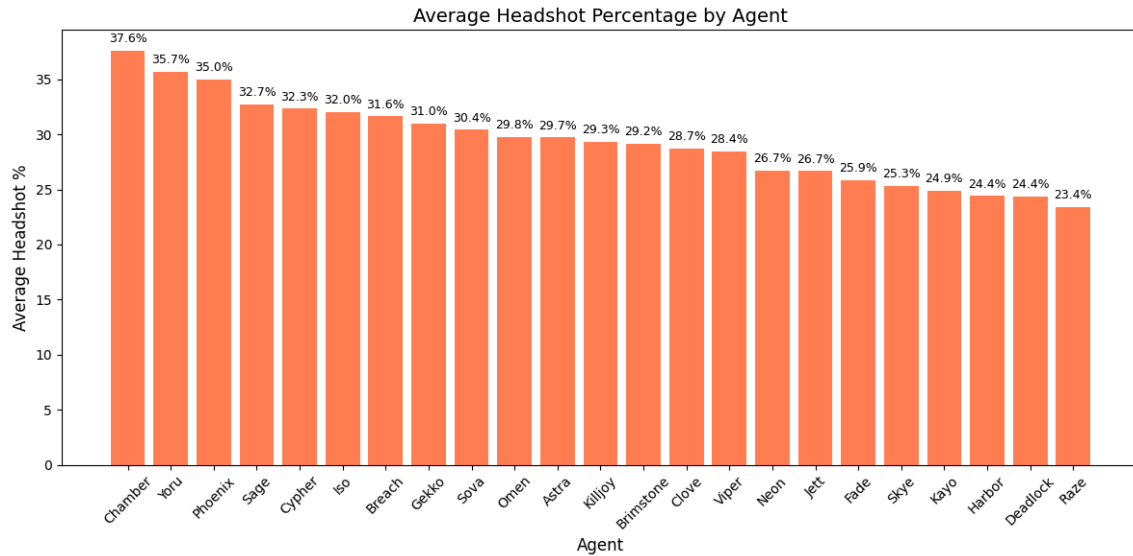
The heatmap and individual map charts together give a clear picture of agent trends across the pool. Omen and Sova stand out as the most consistent anchors, appearing at high utilization across nearly every map. Viper is dominant on maps like Bind and Icebox, while Cypher shows strong presence on Sunset and Haven. Agents like Fade, Raze, and Killjoy are impactful but more map-dependent, with Killjoy peaking on Icebox and Fade/Raze shaping Lotus and Bind.

The heatmap also highlights niche or situational picks — agents such as Neon, Astra, Gekko, and Harbor show up in specific contexts but don’t carry across the pool. This reinforces the idea that while a core set of agents define the meta everywhere, teams flex into specialized choices depending on map layout and strategy.

- Average Kill Contribution by Agent**



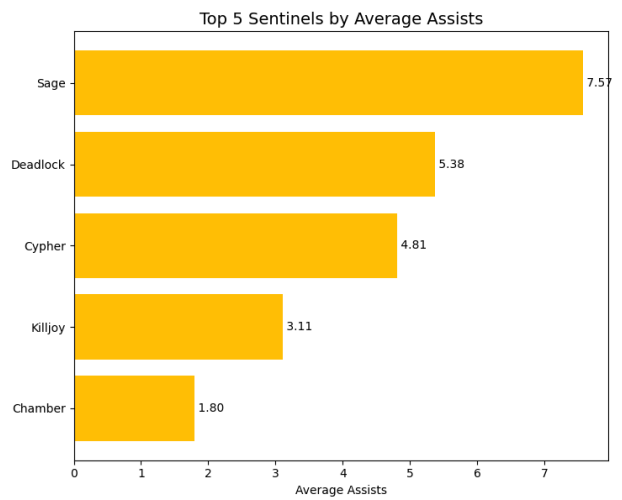
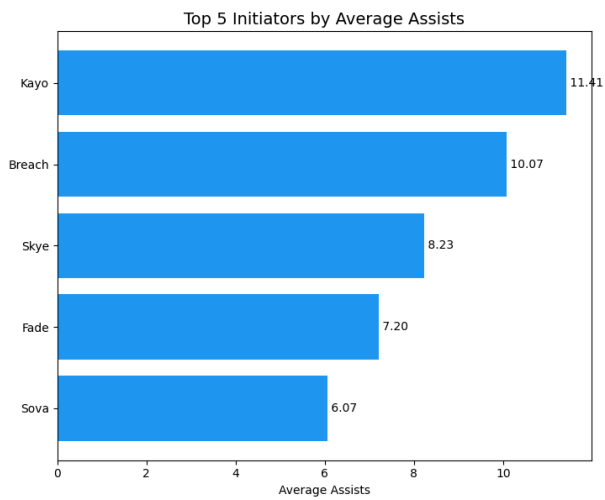
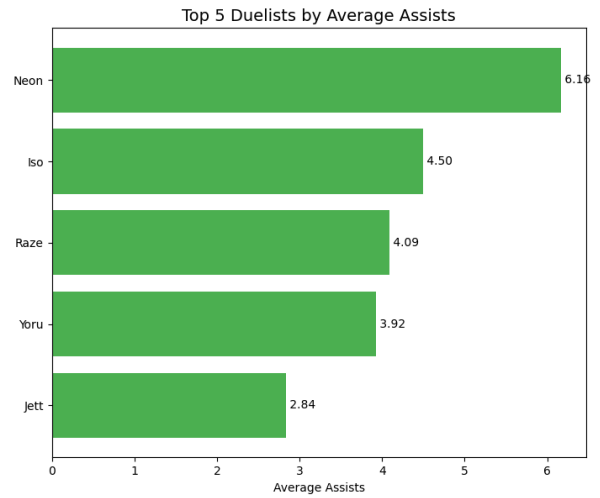
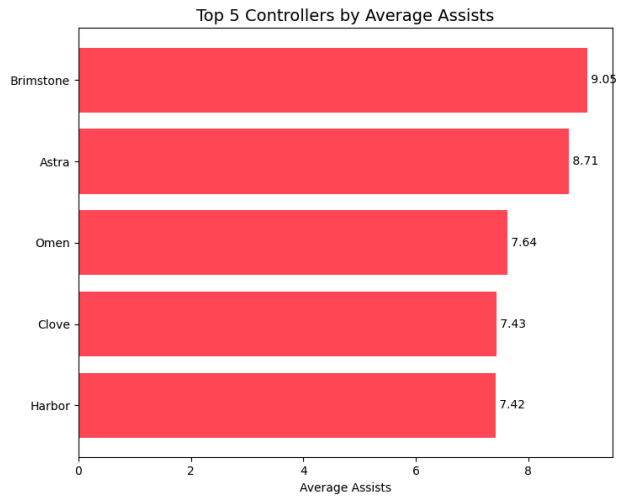
- Average Headshot Percentage By Each Agent**



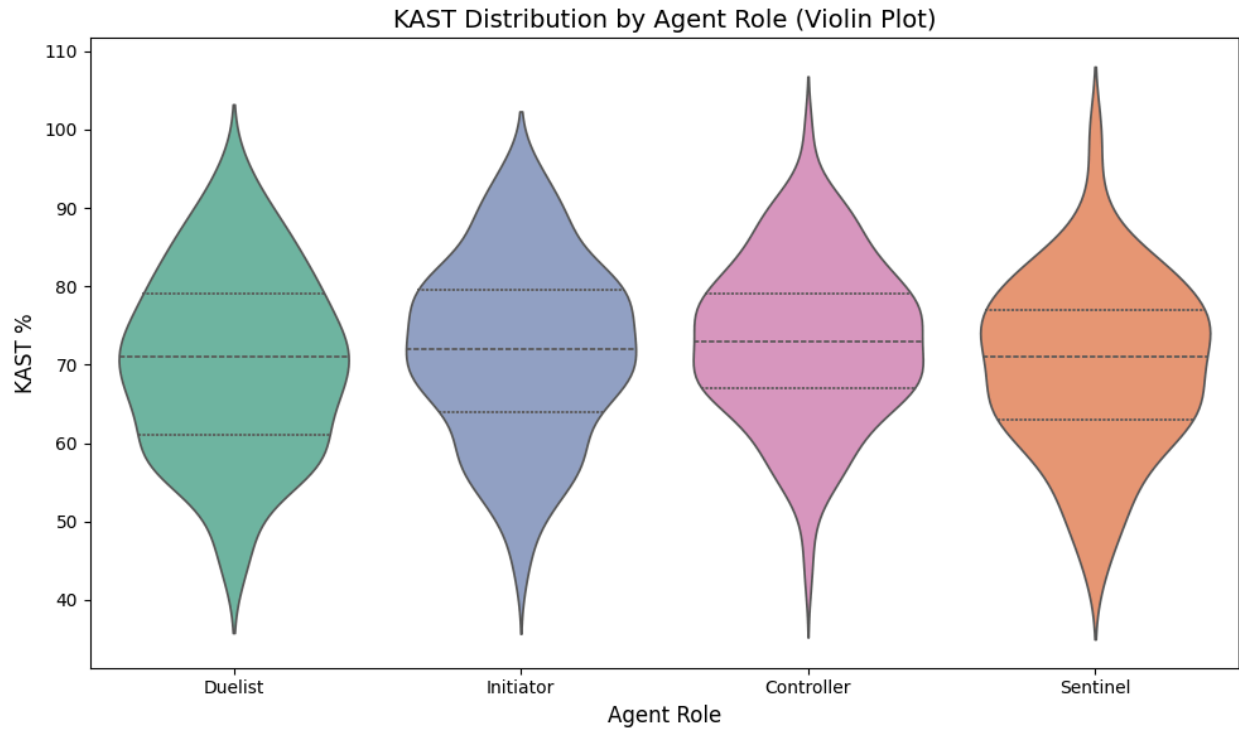
The kill contribution chart highlights fragging leaders like Iso, Neon, Harbor, and Raze, who consistently deliver high kill counts. On the other hand, utility-driven agents such as Skye, Phoenix, and Deadlock contribute fewer kills but bring value through abilities and map control. The headshot percentage chart adds another layer: Chamber, Yoru, and Phoenix stand out for precision, while agents like Raze, Skye, and Harbor trend lower, reflecting different playstyles.

- **Fragging Power:** Maps that demand aggressive entry or duel-heavy play benefit from agents like Iso or Neon, but only if the player assigned can consistently handle high-pressure fragging roles.
- **Accuracy Specialists:** Agents with high headshot percentages (Chamber, Yoru, Phoenix) should be slotted into positions where precision matters most, such as holding long sightlines or taking aggressive fights.
- **Utility Balance:** Lower-frag agents like Skye or Killjoy still play critical roles in map control and information gathering. Coaches need to ensure these picks align with a player's strengths in utility usage rather than raw aim.
- **Player Fit:** Beyond agent stats, coaches must weigh individual player skillsets and weaknesses. A mechanically gifted player might thrive on Iso or Jett, while a more strategic, utility-focused player could excel on Killjoy or Brimstone.

- **Top 5 Agents by Average Assist per role**



- **KAST Distribution by Agent Role**



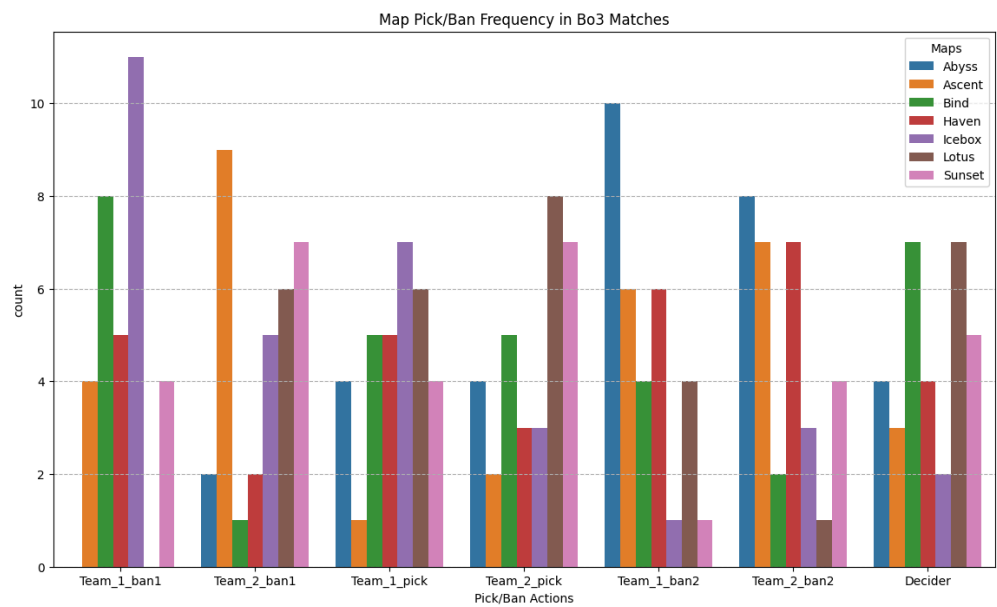
- **Rating Distribution by Agent Role**



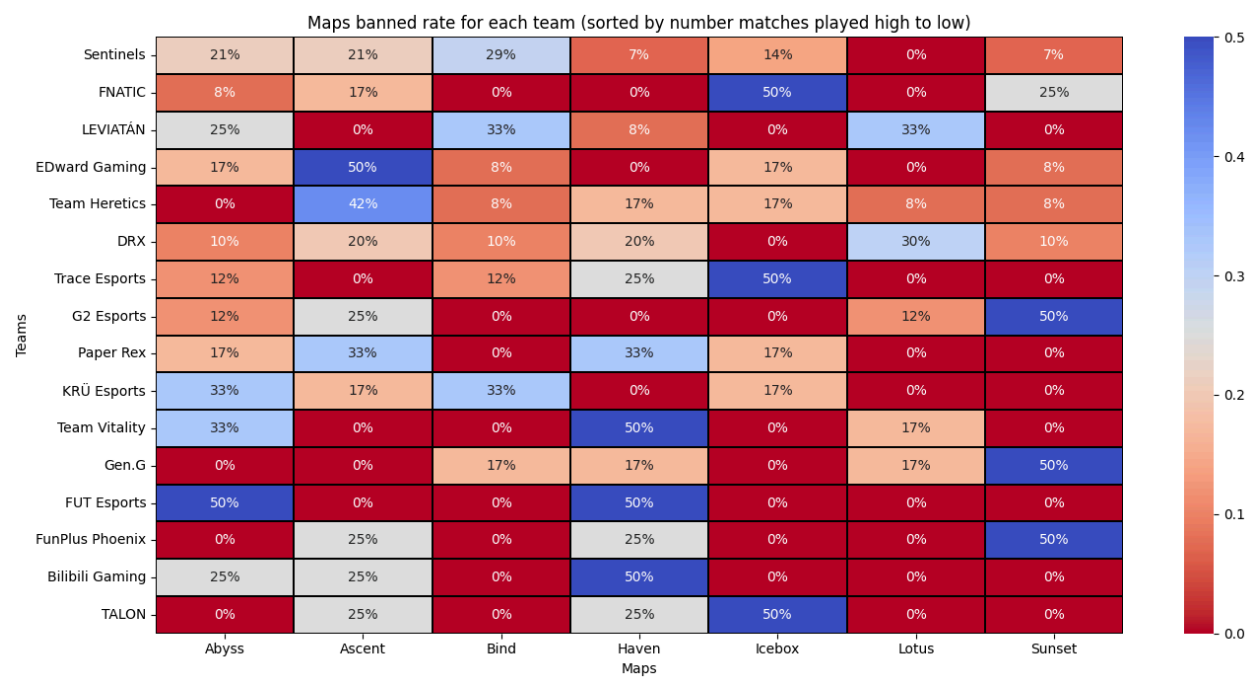
Controllers and Initiators rack up the most assists, showing their value in setting up plays and provide supporting utility for their duelist teammates. Duelists sit lower since their job is more

about fragging than support. KAST and rating spreads confirm it — Duelists are volatile but can swing games, while Controllers/Initiators stay steady. For coaches, the key is matching roles to player strengths: put consistent players on utility roles, and let your high-skill fraggers run Duelists. Balance is what keeps comps stable.

● Map Pick/Ban Frequency in Bo3 Matches



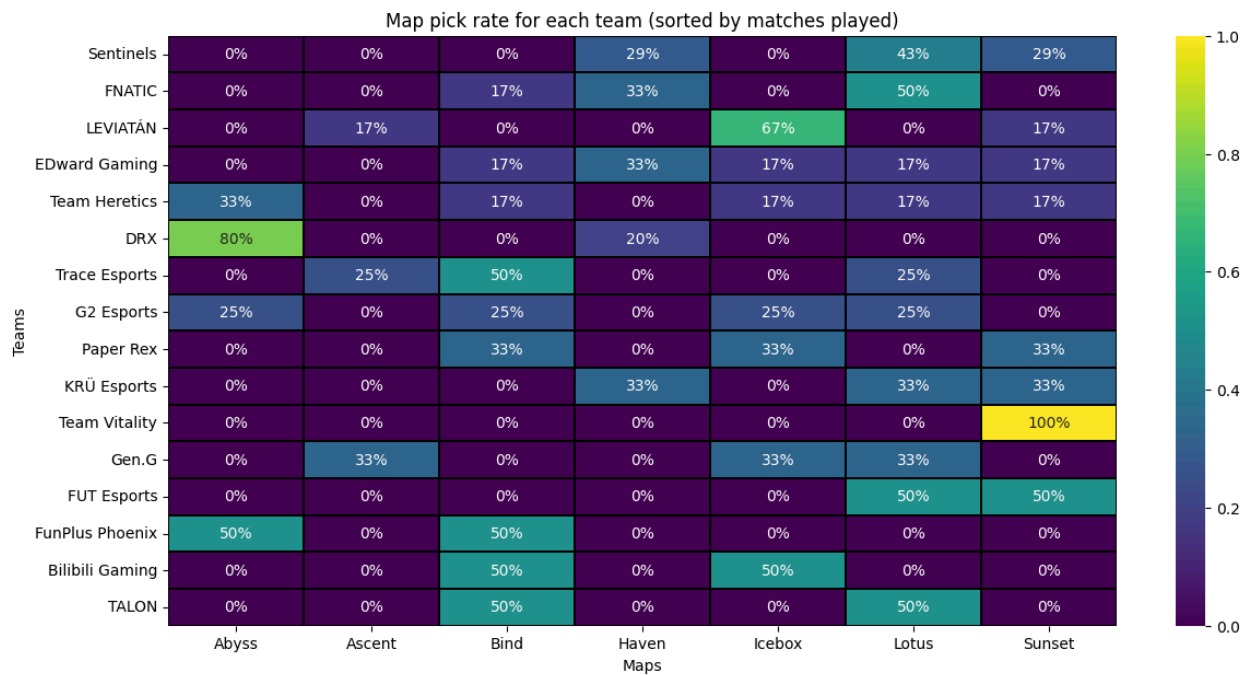
● Maps banned rate for each team on each map (sorted by number matches played high to low)



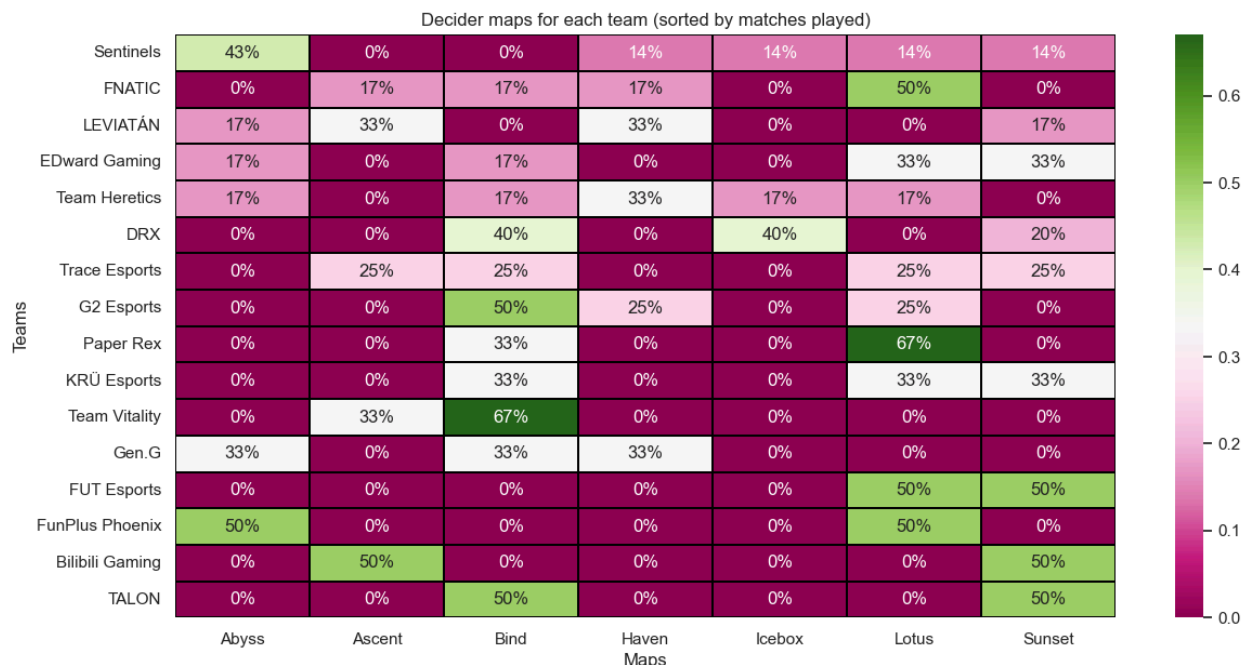
The Bo3 pick/ban chart shows clear preferences: maps like Icebox and Bind are banned more often, while Lotus and Ascent appear frequently as picks or deciders. This suggests that some maps are seen as riskier or harder to master, while others are considered safer choices in competitive play.

The team-level heatmap adds another layer. Certain teams show strong tendencies — for example, Fnatic heavily avoids Icebox, while EDG often bans Ascent. Others spread bans more evenly, which makes their veto strategy harder to predict.

● Map pick rate for each team (sorted by matches played)



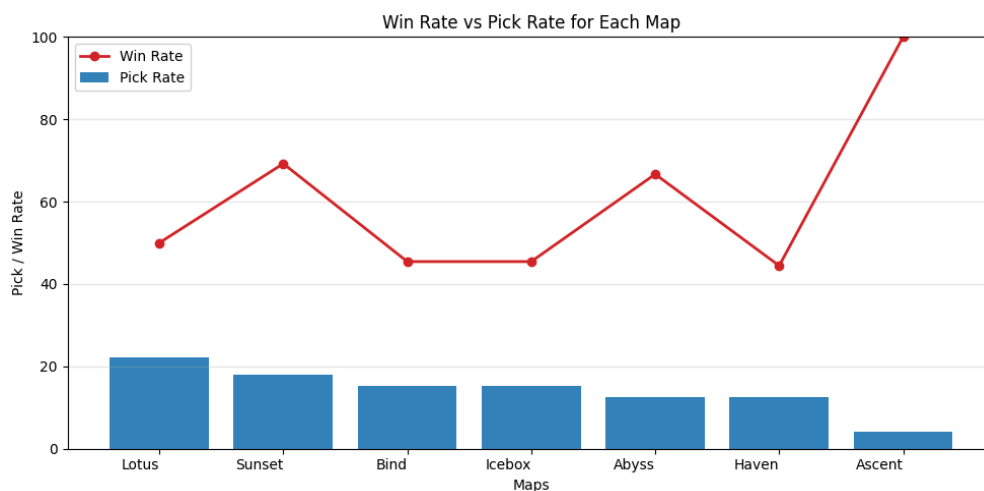
● Decider maps for each team (sorted by matches played)



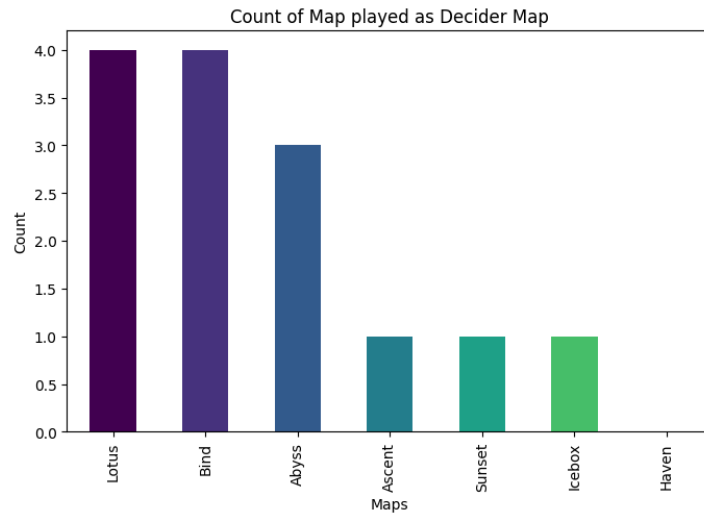
The pick-rate heatmap shows clear comfort maps for certain teams — for example, **DRX heavily favor Abyss**, while **Fnatic lean on Lotus**. Others spread their picks more evenly, like EDG and G2, which makes their pool harder to predict.

The decider map chart adds another layer: some teams consistently end up on the same deciders (e.g., **Sentinels often on Abyss**, **Vitality on Bind**), while others rotate more. This highlights which maps are trusted fallback options versus those avoided in high-pressure situations.

- Overall Pick Rate vs Win Rate for each map

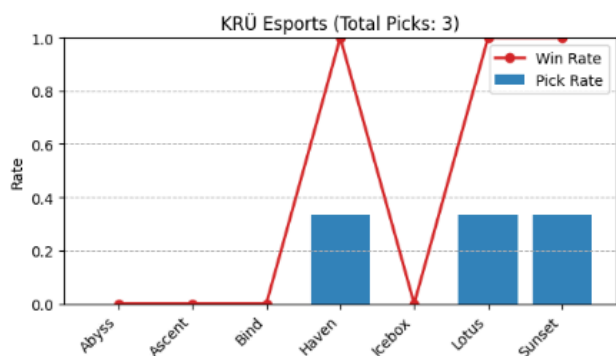
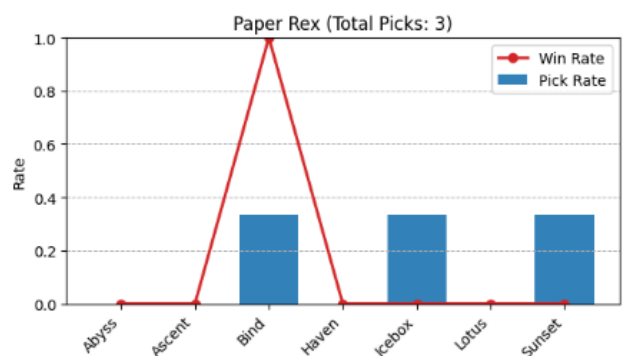
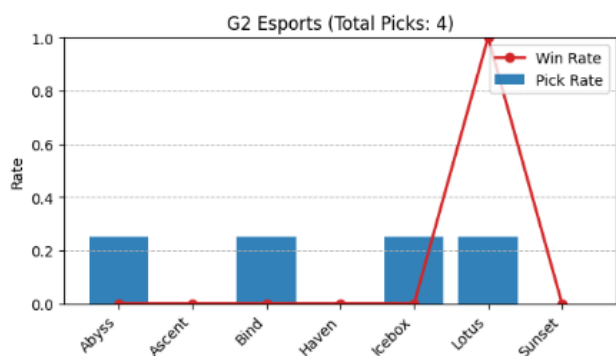
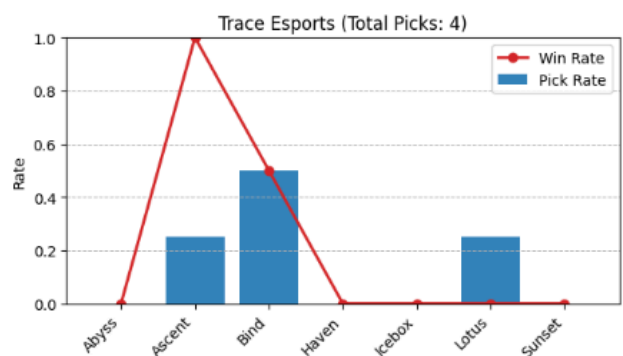
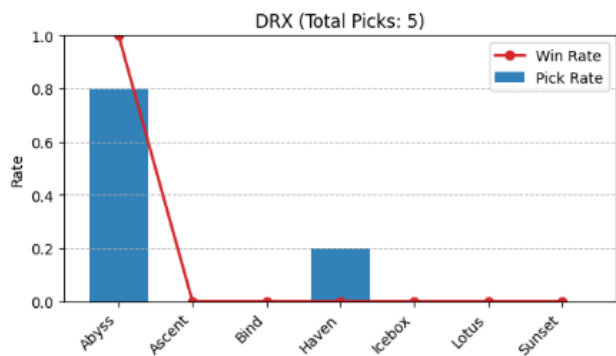
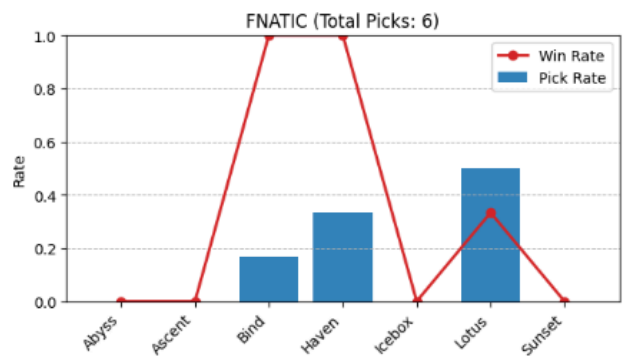
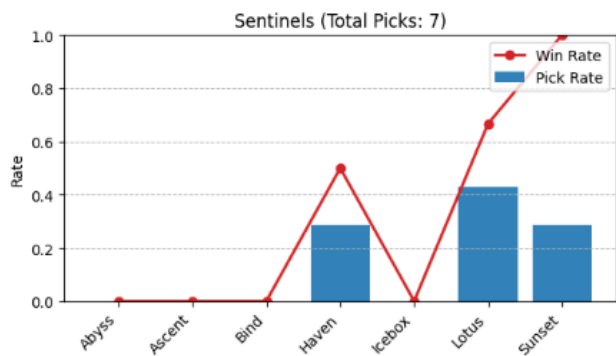
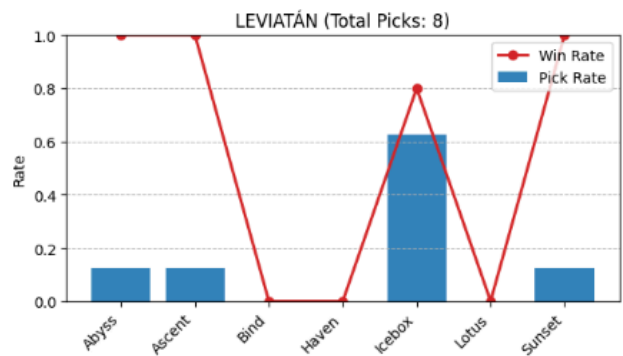
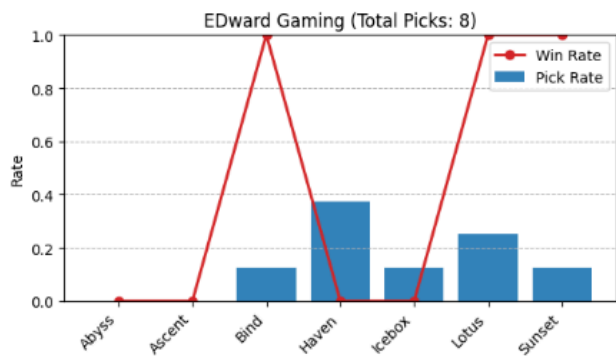
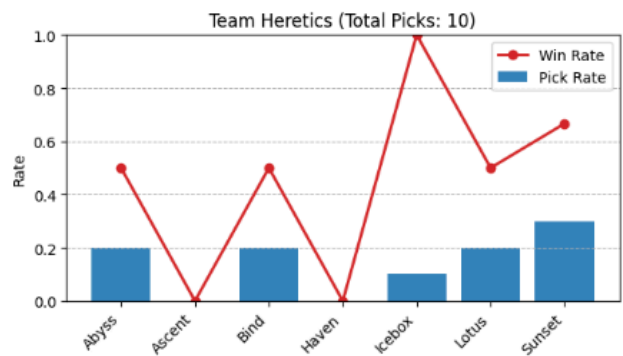


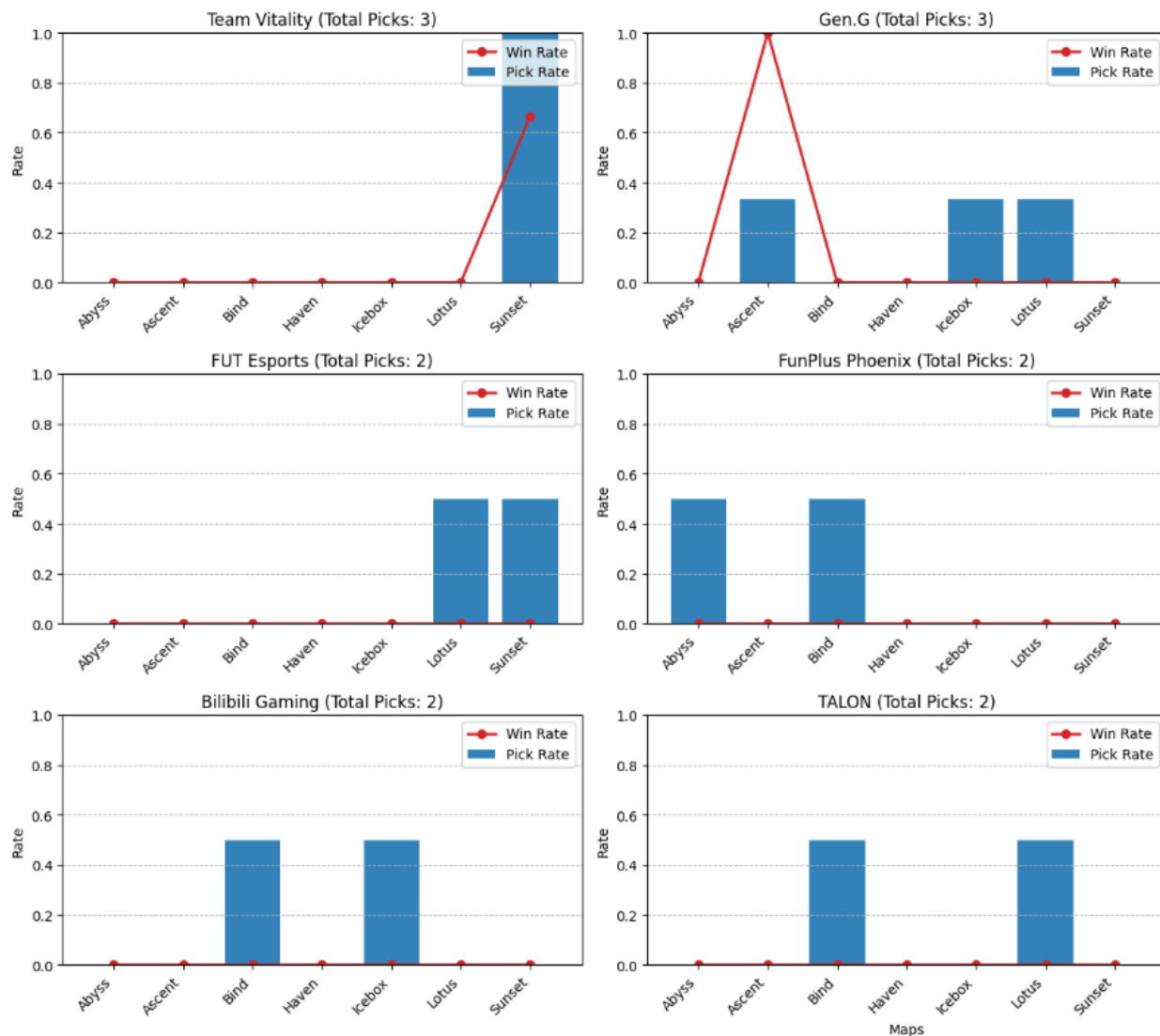
- Number of times each map was played as decider



The win rate vs pick rate chart shows a clear split: Ascent sits at 100% win rate but is barely picked, while Lotus is the most played yet only hovers around 50% wins. Sunset and Abyss stand out as efficient maps — solid win rates despite lower pick frequency. Bind, Icebox, and Haven sit in the middle with balanced but unspectacular numbers. The decider map chart adds context: Lotus and Bind are the most common deciders, with Abyss also appearing often. Maps like Haven rarely show up as deciders, suggesting teams avoid them in high-pressure situations.

- **Map pick rate vs win rate for each team**





The chart shows clear differences between top teams with larger match counts and lower teams with smaller pools.

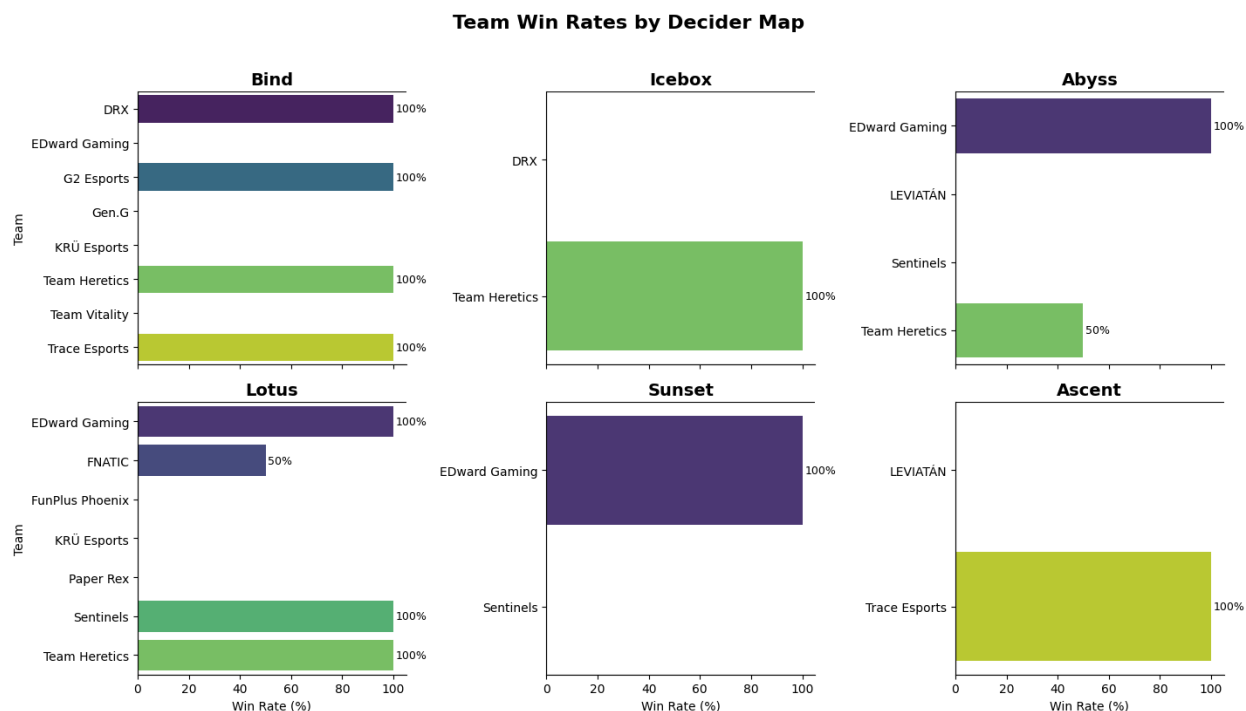
Top teams (Heretics, EDG, Leviatán, Sentinels, Fnatic): They spread picks across multiple maps, but each has clear preferences — Fnatic lean heavily on Lotus, Leviatán on Icebox, DRX on Abyss, while Sentinels split between Lotus and Sunset. These teams show more diversity overall, though not always dominance on every map.

Mid-tier teams: They often rely on one or two comfort maps (e.g., Trace on Bind, DRX on Abyss) which makes them predictable but stable.

Lower teams: With fewer matches, their pools are much narrower — some even specialize almost entirely on one map (Vitality on Sunset, FPX on Abyss/Bind). This gives them sharper comfort but less flexibility.

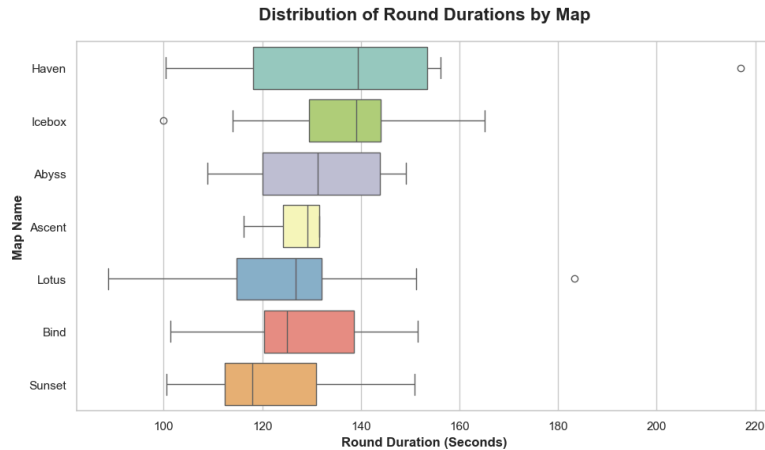
Map Trends: Lotus stands out as the most common pick among top teams, while Icebox splits opinions with Leviatán favoring it and Fnatic avoiding it. DRX rely heavily on Abyss, though most teams use it sparingly. Bind serves as a reliable fallback for mid and lower teams, and Sunset remains a niche choice trusted by only a few like Sentinels and Vitality.

- **Team Win Rates on Decider Maps**



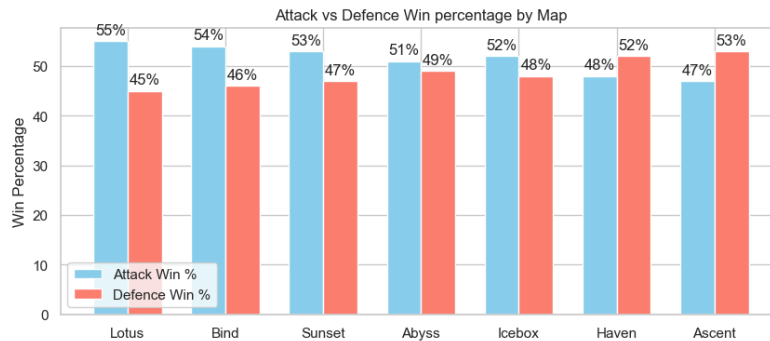
The decider map data shows clear differences in reliability across the pool. Bind emerges as the most consistent decider, with several top teams (DRX, EDG, G2, Heretics, Trace) all holding perfect win records when it appears, making it the safest fallback option. Lotus is popular but mixed, with some teams flawless on it while others split results, showing it's a comfort map but not universally dominant. Abyss and Sunset appear less often but deliver strong outcomes for the teams that trust them, while Icebox and Ascent are rare choices, only converting into wins for select squads like Heretics and Trace.

- **Distribution of Round Duration by Map**



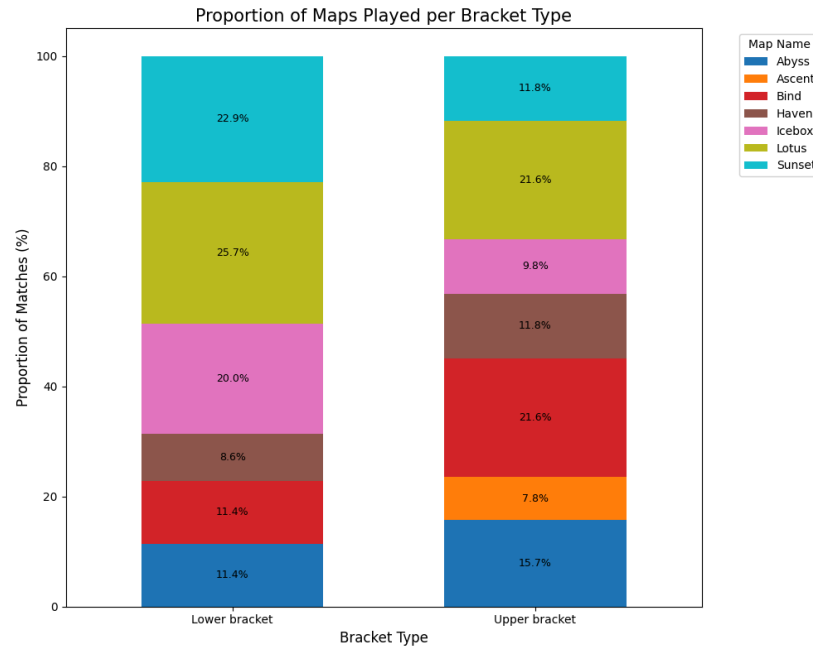
The distribution shows noticeable differences in pacing across maps. Abyss and Ascent trend toward longer rounds, with wider spreads suggesting slower, more tactical play. Icebox and Haven sit in the middle, offering balanced timings. Lotus and Bind lean shorter, reflecting faster engagements and decisive rounds. Sunset shows the tightest distribution, indicating more consistent round lengths compared to other maps.

- **Attack vs Defence Win Percentage for each Map**



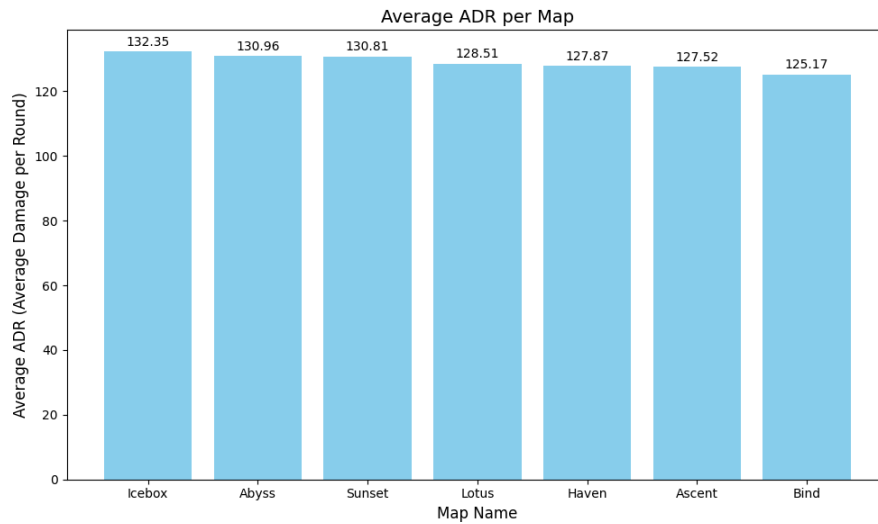
Side win rates reveal a slight tilt toward attackers on most maps — Lotus, Bind, Sunset, Abyss, and Icebox all favor attacking sides, with win percentages around 51–55%. In contrast, Haven and Ascent tilt toward defense, giving defenders a small edge (52–53%). Most maps being attack favoured implies that teams mostly go for set attack plan or strats

- **Proportion of Maps Played per Bracket Type**

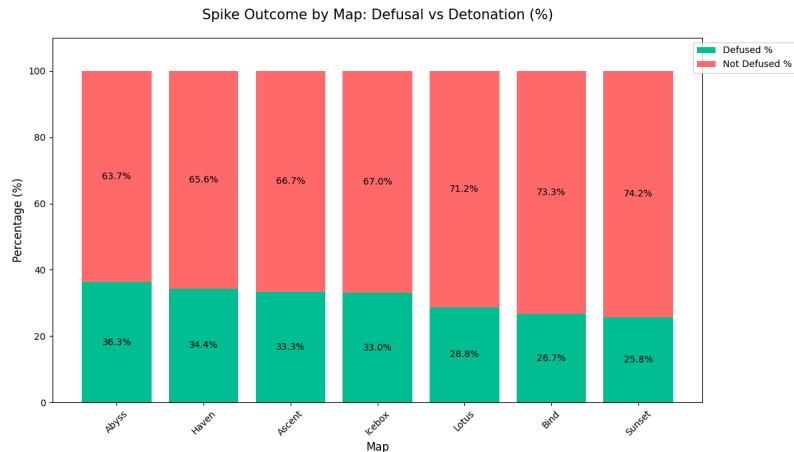


The distribution shows that Lotus and Sunset dominate lower bracket matches, while Bind and Abyss appear more often in upper brackets.

- **Average ADR per Map**



- **Spike Outcome By Map : Defusal vs Detonation (%)**

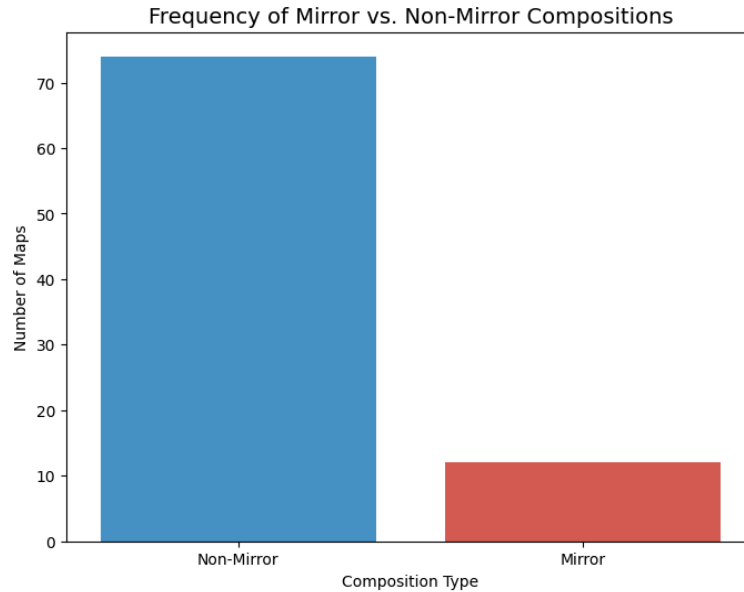


ADR by Map: Damage output peaks on Icebox (132.35), with Abyss and Sunset close behind (~131). Lotus, Haven, and Ascent sit mid-range (~127–128), while Bind records the lowest ADR (125.17). This indicates that maps like Icebox and Abyss encourage heavier firefights, while Bind tends to produce lower-damage, slower rounds.

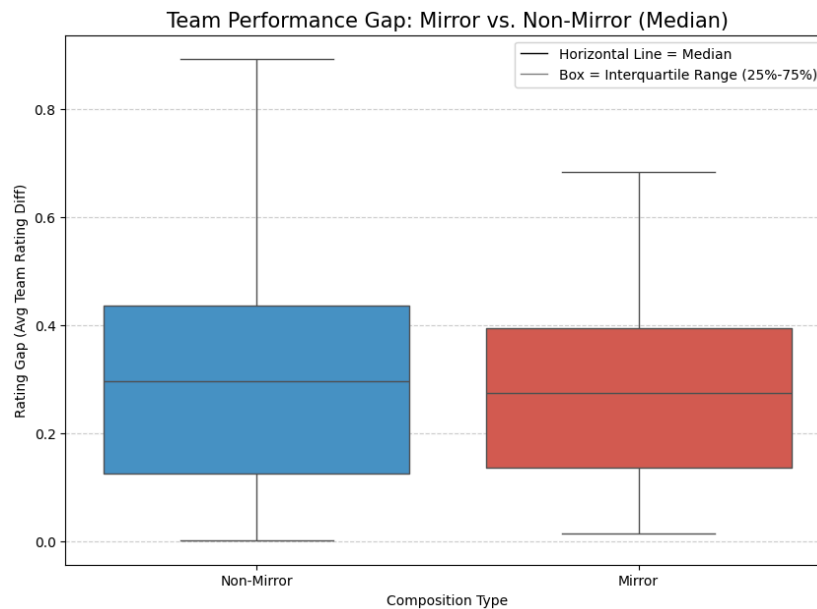
Spike Outcomes: Across all maps, spike detonations outweigh defusals. Bind and Sunset have the lowest defusal rates (~26–25%), meaning spikes planted here are very likely to go off. Lotus also leans heavily toward detonation (71%), while Abyss, Haven, Ascent, and Icebox show slightly higher defusal chances (~33–36%) but still favor detonation overall.

Strategic Implications: Teams can use this data to adapt their approach depending on the map's tendencies. On maps with high detonation rates (Bind, Sunset, Lotus), defenders may need to focus more on denying the plant altogether, since post-plant retakes are statistically less successful. Conversely, on maps with higher defusal chances (Abyss, Haven, Ascent, Icebox), defenders can afford to let the spike go down and then play for coordinated retakes. Attackers, meanwhile, can plan around whether their strength lies in securing plants quickly or holding post-plant setups under pressure.

- **Frequency of Mirror vs Non- Mirror Compositions**

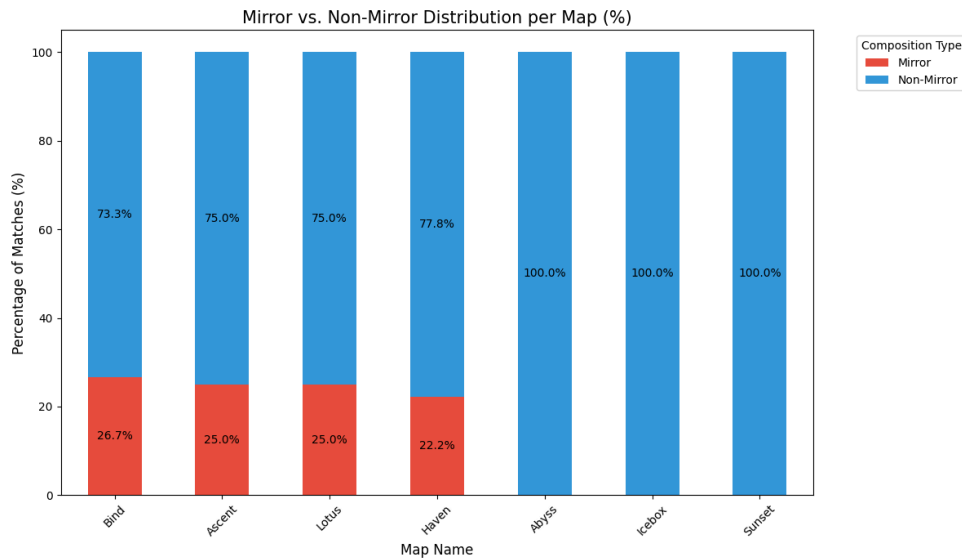


- **Team Performance Gap : Mirror vs Non-Mirror**



Non-Mirror compositions dominate competitive play, appearing far more frequently than Mirror setups, which shows teams prefer variety and asymmetry in their strategies; however, the performance data reveals that Mirror compositions generally produce smaller rating gaps and more balanced matches, while Non-Mirror setups tend to amplify differences in team strength or comfort, meaning that although teams overwhelmingly choose diverse, non-mirrored approaches for flexibility and surprise, these choices often lead to greater disparities in outcomes compared to the tighter, evenly matched games seen with mirrored lineups.

- **Mirror Vs Non-Mirror Composition Distribution per Map**



Across the map pool, Non-Mirror compositions dominate, with maps like Abyss, Icebox, and Sunset showing 100% Non-Mirror usage, while Bind, Ascent, Lotus, and Haven still lean heavily toward Non-Mirror (73–78%) but allow for occasional Mirror setups. This reinforces the overall trend that teams overwhelmingly prefer diverse, asymmetric strategies.

7.3 Key Findings

- A small core agent pool defines the meta, but map context drives final decisions.**
 Agents like Omen and Sova form the backbone across nearly all maps, while picks such as Viper, Cypher, Fade, and Raze gain value only on specific map types. Teams consistently balance meta stability with situational flexibility rather than forcing uniform compositions.
- Agent effectiveness is role-dependent, not purely frag-driven.**
 Duelists deliver high but volatile impact through kills, while Controllers and Initiators provide steadier value via assists, KAST, and round control. Successful teams assign agents based on player strengths and role fit rather than raw pick rates.
- Map veto and pick strategies reflect risk management more than popularity.**
 Frequently banned maps like Icebox and Bind indicate higher execution complexity, while Lotus and Ascent are trusted as safer picks or deciders. Teams with broader and more balanced map pools are less predictable and better positioned in Bo3 formats.
- Decider maps function as strategic safety nets.**
 Bind emerges as the most reliable decider across multiple teams, while Lotus appears frequently but delivers mixed outcomes. Rare deciders such as Haven and Ascent suggest teams avoid them in high-pressure scenarios.
- Map structure directly shapes pacing, damage output, and post-plant success.**
 Icebox and Abyss encourage longer rounds and higher ADR, whereas Bind and Lotus

favor faster executions. Maps with high detonation rates require stronger plant-denial strategies, while others allow defenders to rely more on coordinated retakes.

- **Non-Mirror compositions dominate, but come with higher outcome variance.**
Teams overwhelmingly favor asymmetric compositions for flexibility and surprise, yet Mirror matchups produce closer, more balanced games. This highlights a strategic trade-off between creative freedom and consistency.
- **Top teams succeed through adaptability rather than strict meta imitation.**
Strong teams maintain flexible map pools, adjust compositions based on opponent and context, and exploit agent–map synergies, while weaker teams rely heavily on limited comfort picks.

8. Pillar 2: Player Performance & Efficiency

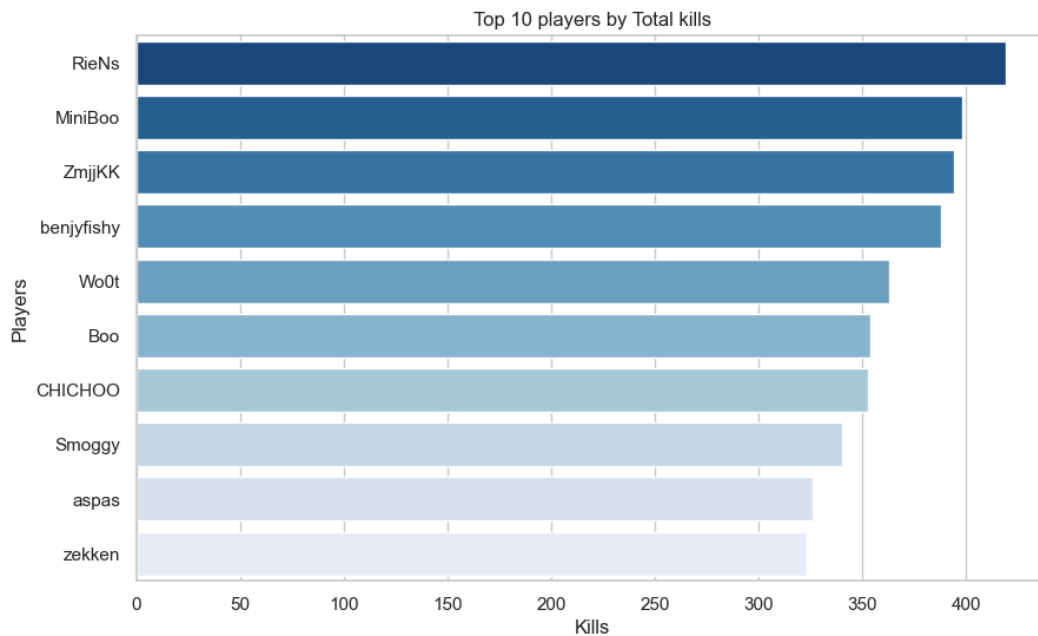
(Detailed_player_stats.ipynb, Player_stats.ipynb)

8.1 Analysis Focus

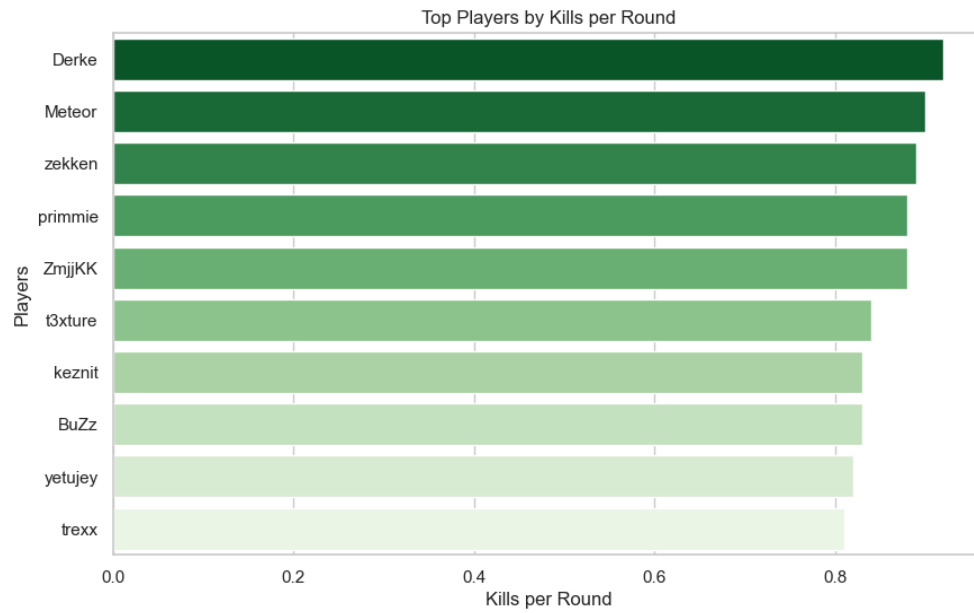
- Player fragging output and efficiency (kills, ACS, ADR)
- Performance consistency across roles and match outcomes (rating, KAST, K/D)
- Impact trade-offs between fragging and support play (assists, first kills, first deaths)
- Clutch performance and high-pressure decision-making (1vX wins, clutch %)
- Objective contribution and round conversion (plants, defuses)
- Multi-kill impact and round-swing potential (2K–5K distribution)
- Overall player and team effectiveness using composite performance scores

8.2 Analysis

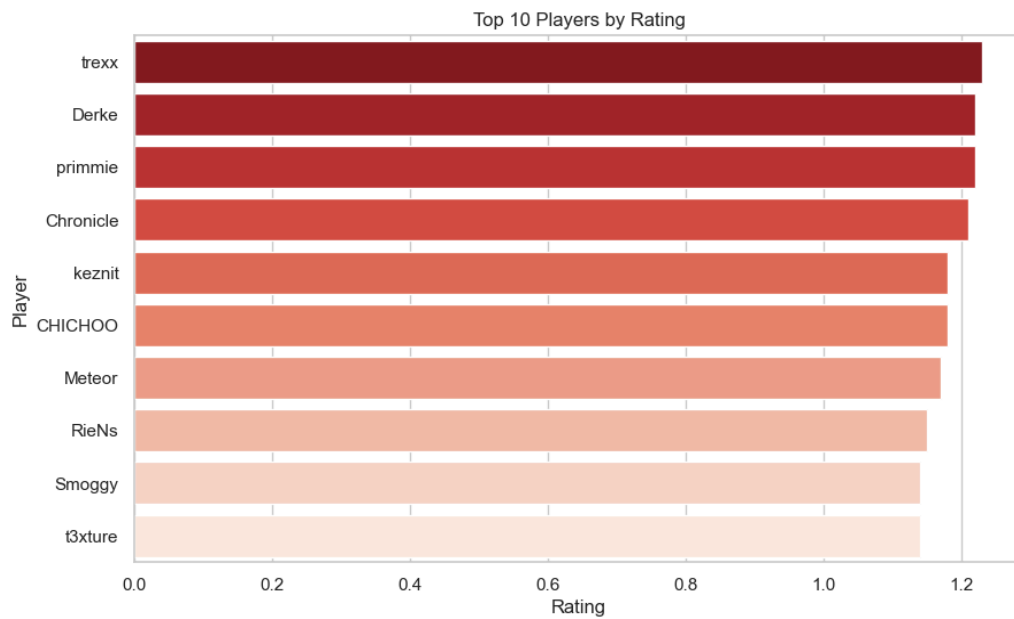
- **Top 10 Players by Total Kills**



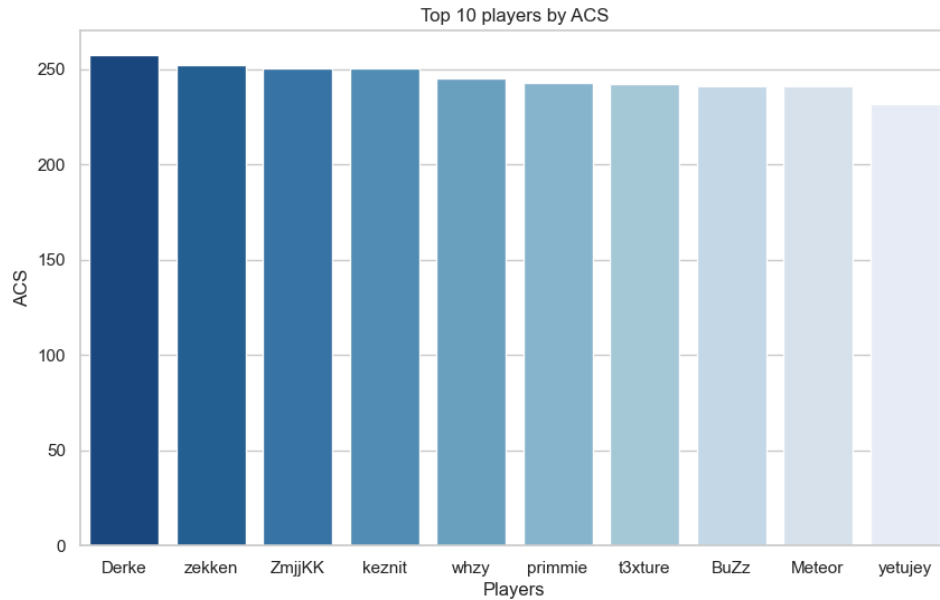
- **Top 10 Players by Kills per Round**



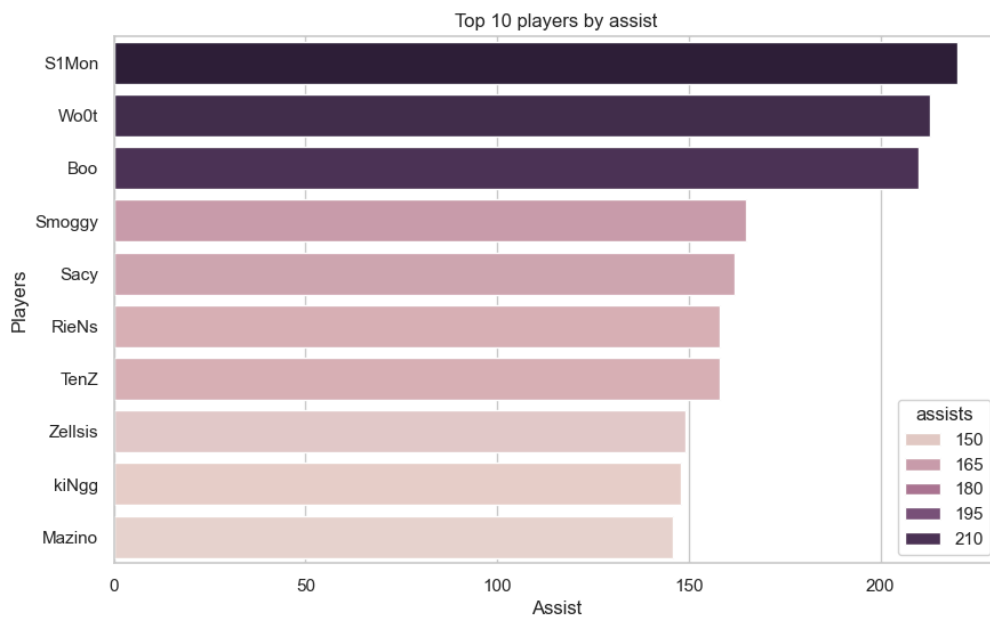
- **Top 10 Player by Rating**



- **Top 10 Players by ACS**



- **Top 10 players by assist**

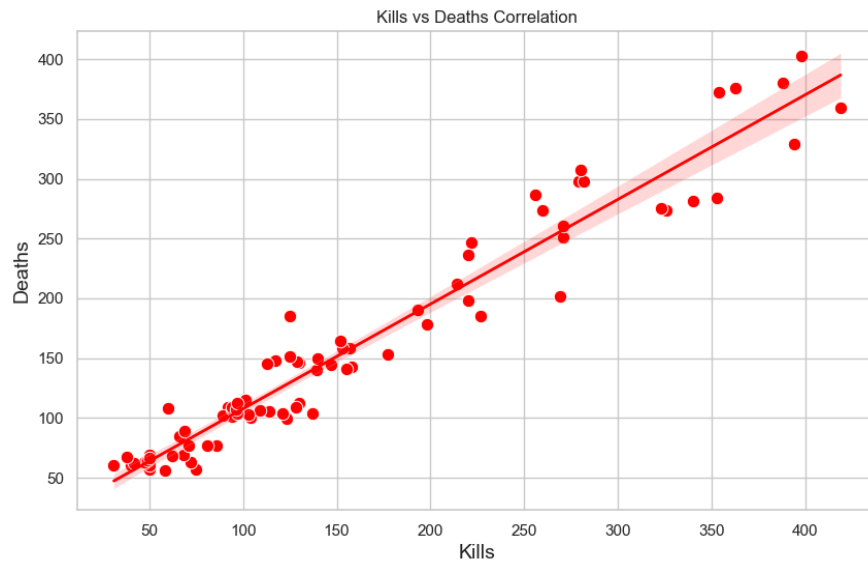


Kills per Round & ACS: Players like Derke, zekken, and ZmjjKK dominate both kills per round and ACS, showing they consistently convert fights into high-impact rounds. These are the pure fraggers who set the pace.

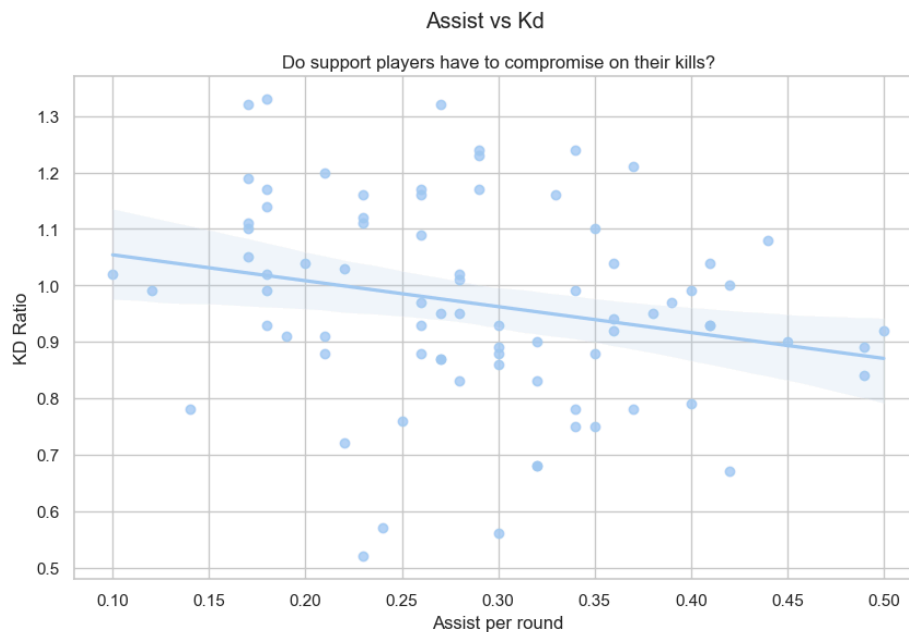
Ratings: Interestingly, trexx tops the rating chart, despite not leading in kills or ACS. This suggests efficiency, consistency, and impact beyond raw fragging — a player who maximizes value in every round. Others like Chronicle and CHICHO0 also appear here, highlighting supportive or hybrid roles that still achieve high ratings.

Assists: On the other end, players like S1Mon, Wo0t, and Boo lead in assists, showing their contribution comes from enabling teammates rather than direct fragging. This balances the frag-heavy stars with utility and teamplay.

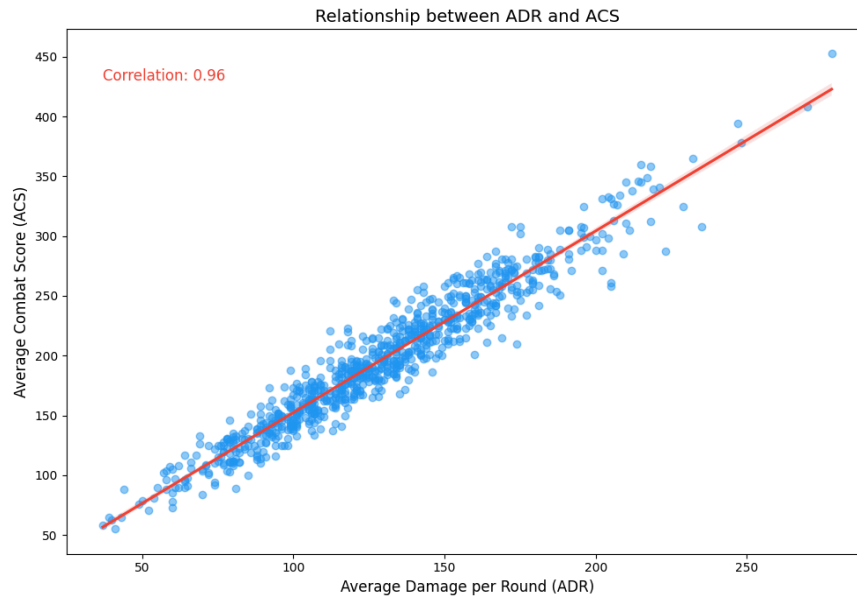
- **Kills vs Death Correlation**



- **Assist vs KD (kill-to-death ratio)**



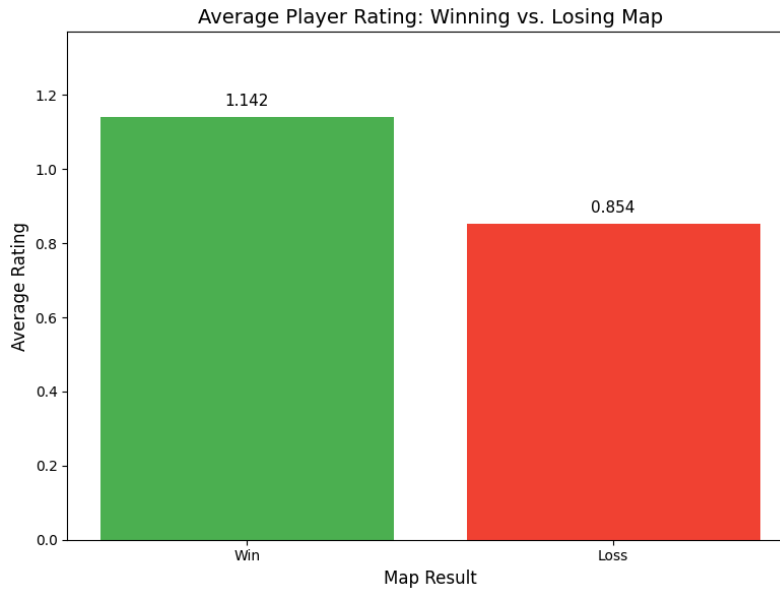
- **Relationship between ADR and ACS**



Looking at the player impact correlations, two clear dynamics emerge. The kills vs deaths scatter shows a positive correlation — players who rack up more kills also tend to die more, reflecting aggressive playstyles that trade risk for impact. Meanwhile, the assists vs KD plot highlights a negative correlation — support players who generate more assists often carry lower KD ratios, suggesting they sacrifice fragging power to enable teammates. Finally, the ADR vs ACS relationship is almost perfectly linear (correlation 0.96), confirming that damage output directly drives combat score, making ADR a reliable predictor of overall impact.

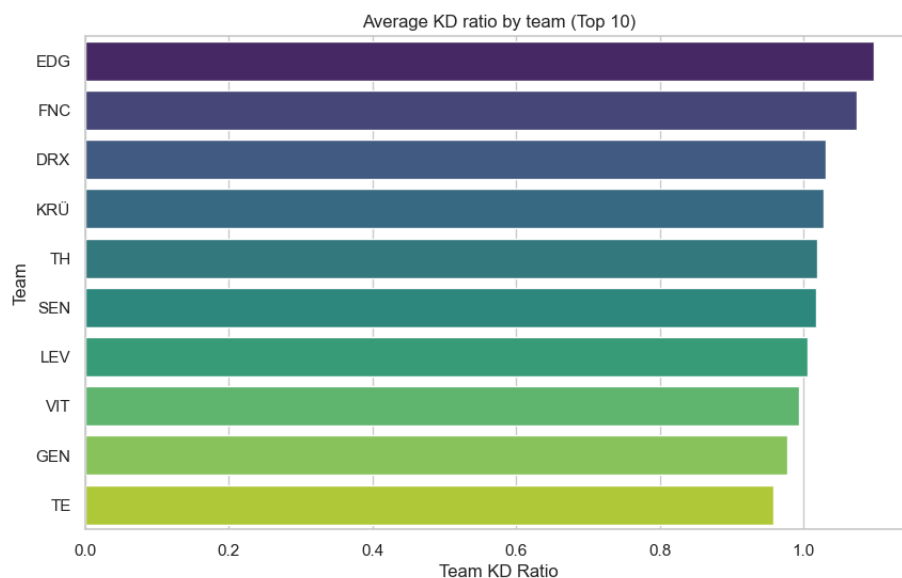
In summary: fraggers push both kills and deaths upward, supports trade KD for assists, and ADR cleanly translates into ACS. Together, these charts show how different roles shape performance — duelists thrive on high-risk, high-reward fragging, support enabled at the cost of KD, and damage remains the most consistent metric of individual impact.

- **Average Player Rating on Winning vs Losing the Map**

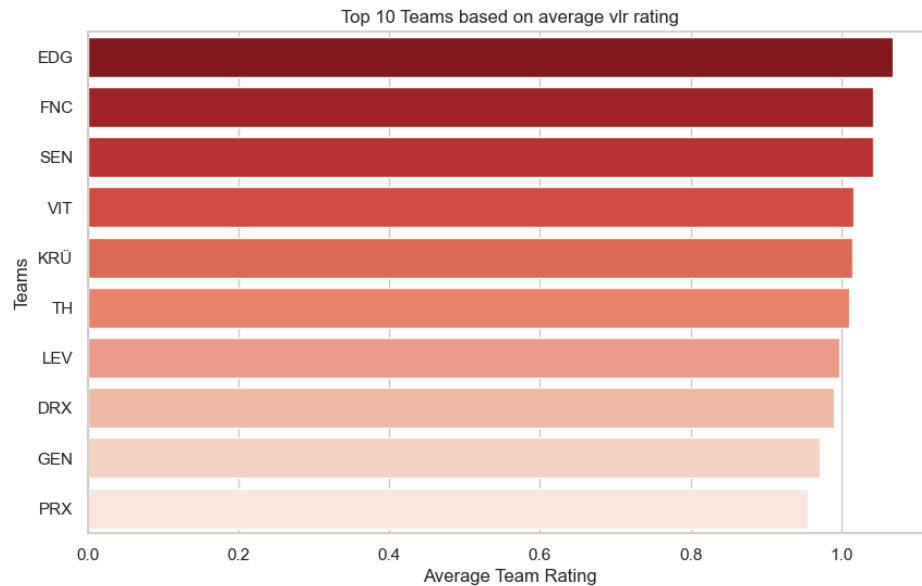


The data shows a clear performance gap: players average a 1.142 rating on winning maps compared to 0.854 on losses, underscoring how individual impact strongly correlates with team success. This aligns with earlier trends — fraggers like Derke and zekken drive high kills and ACS, supports trade KD for assists, and ADR cleanly translates into ACS. Together, the charts highlight that while roles differ in how they contribute, sustained individual efficiency is the decisive factor in converting maps into wins. Teams that balance fragging power, supportive utility, and consistent ratings are best positioned to tilt outcomes in their favor.

- **Average KD ratio by team**

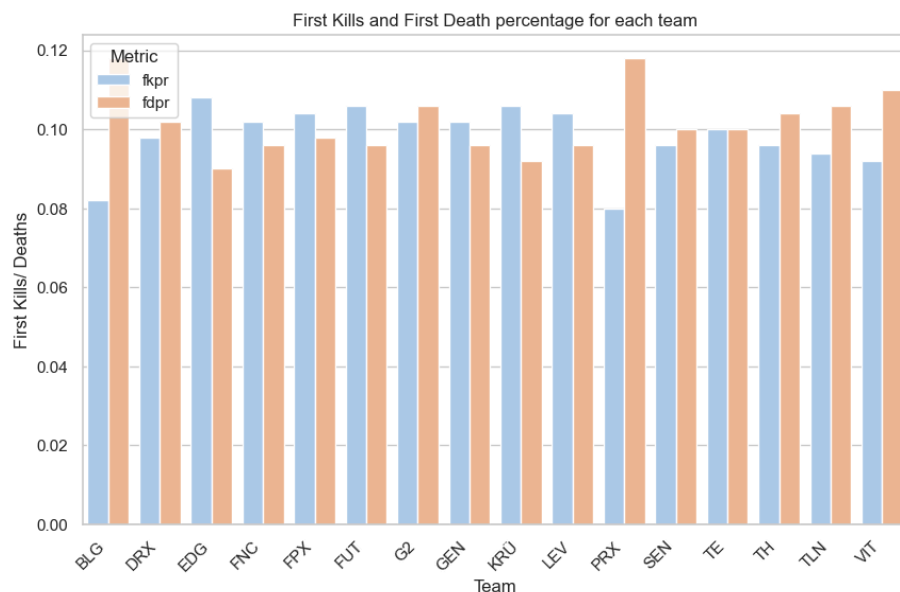


- **Top 10 teams based on the VLR rating**



EDG lead both metrics, topping average KD ratio and vlr rating, showing strong fragging efficiency paired with consistent overall performance. FNC and SEN also rank highly in ratings, while DRX and KRÜ appear in the KD chart but sit lower in ratings, suggesting fragging strength without the same consistency. TE round out the KD list at the bottom, reflecting weaker kill/death efficiency compared to peers.

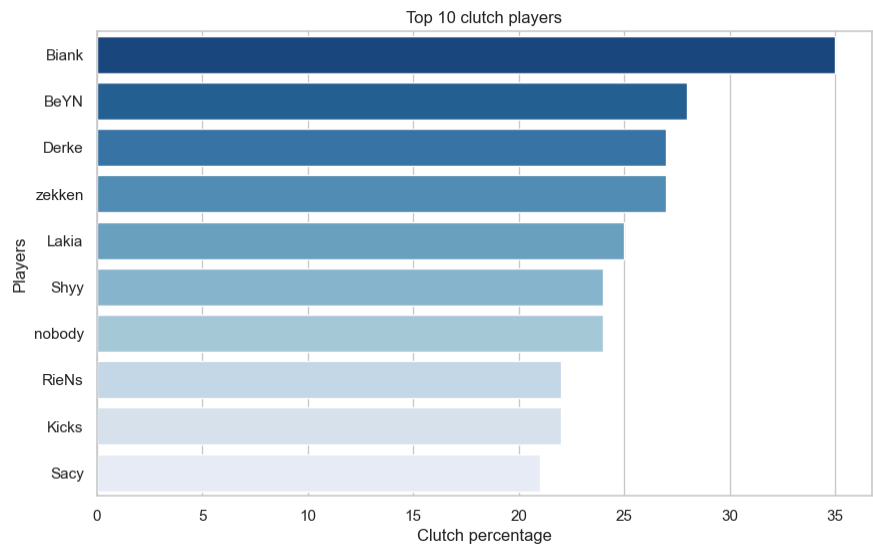
- **First Kills vs First Death percentage for each team**



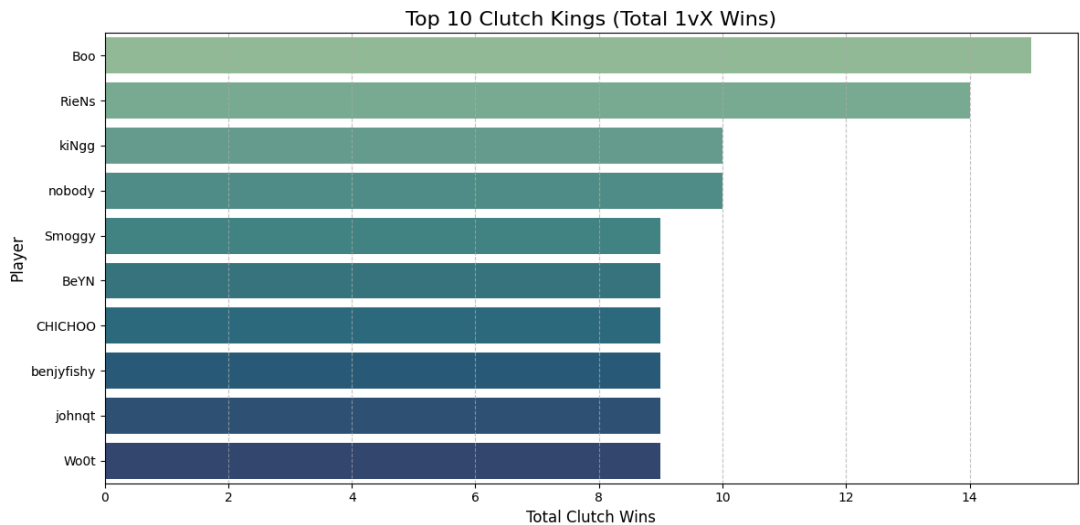
Early-round engagements highlight different team identities. EDG, FNC, and DRX generate high first kill percentages, often setting the tone aggressively. Others like GEN and LEV show higher

first death rates, indicating riskier openings or weaker early setups. Teams like SEN and VIT balance both sides more evenly, reflecting measured approaches to early fights.

- **Top 10 clutch players**

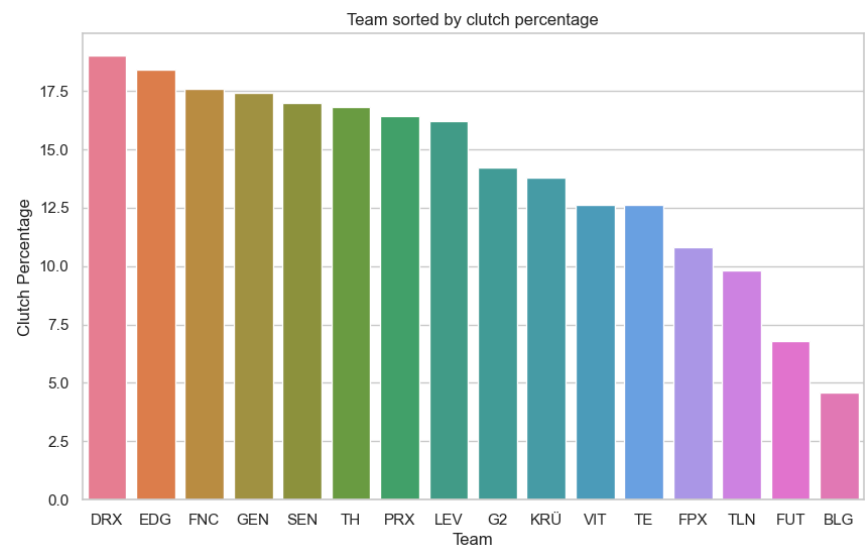


- **Top 10 Clutch Kings (Total 1vX Wins)**



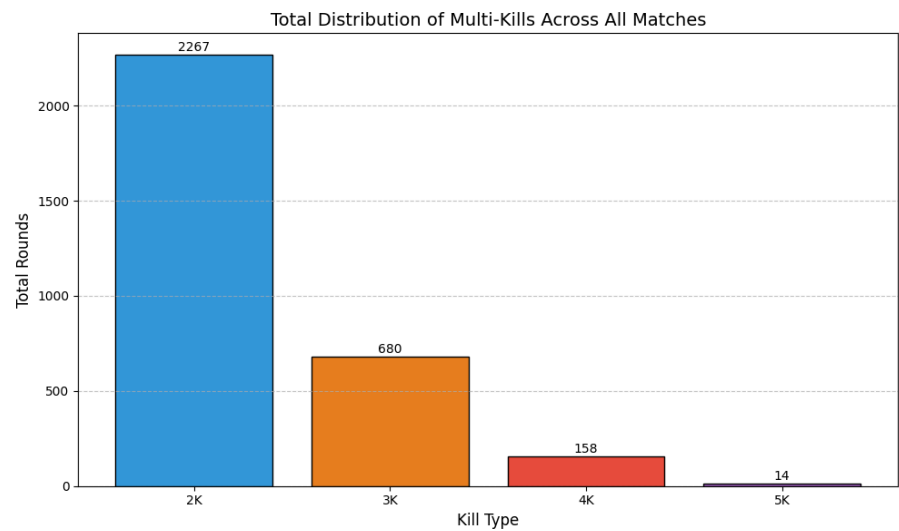
Players like Biank and BeYN lead with ~35% clutch success, showing top efficiency under pressure, while fraggers like Derke and zekken also shine in clutch scenarios. In contrast, Boo (15 wins) and RieNs (14 wins) dominate in total 1vX conversions, proving repeated reliability. Overall, clutch % reflects efficiency, 1vX wins show volume, and since most top clutch kings come from teams with more matches, clutch percentage is the stronger metric to track true under-pressure performance.

- Teams sorted by Clutch percentages

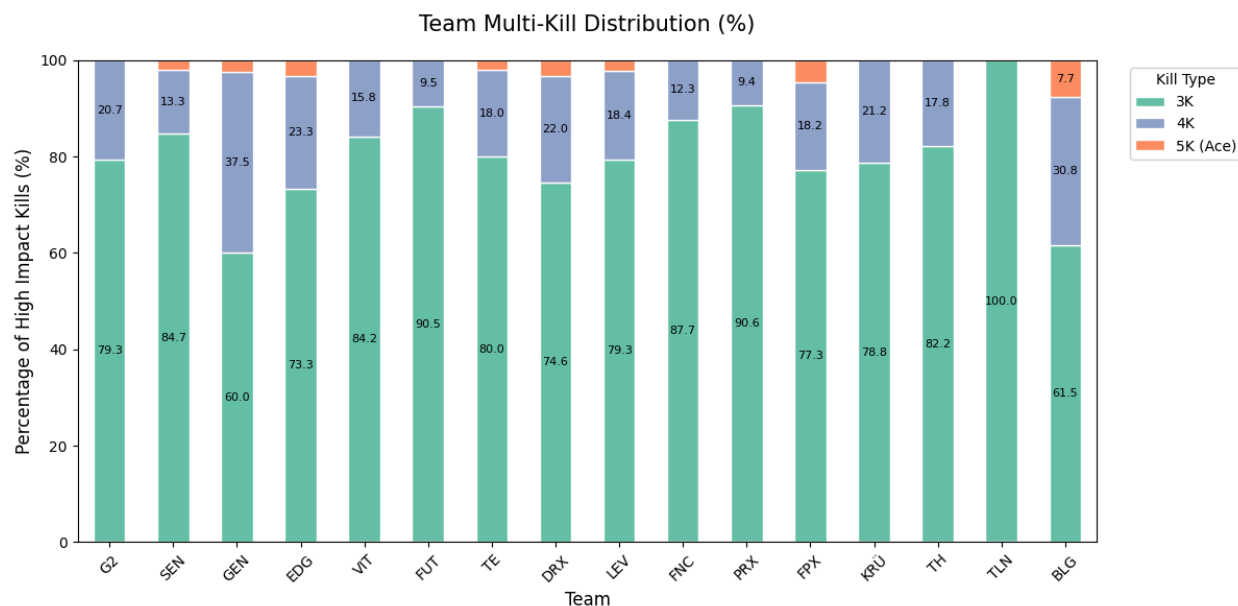


The chart shows DRX leading all teams in clutch percentage, followed closely by EDG and FNC, highlighting their ability to consistently convert high-pressure rounds. Teams like GEN and SEN sit in the mid-range, reflecting balanced but less dominant clutch efficiency. At the lower end, BLG recorded the weakest clutch percentage, suggesting struggles in closing out tight scenarios.

- Total distribution of Multi-Kills Across All Matches

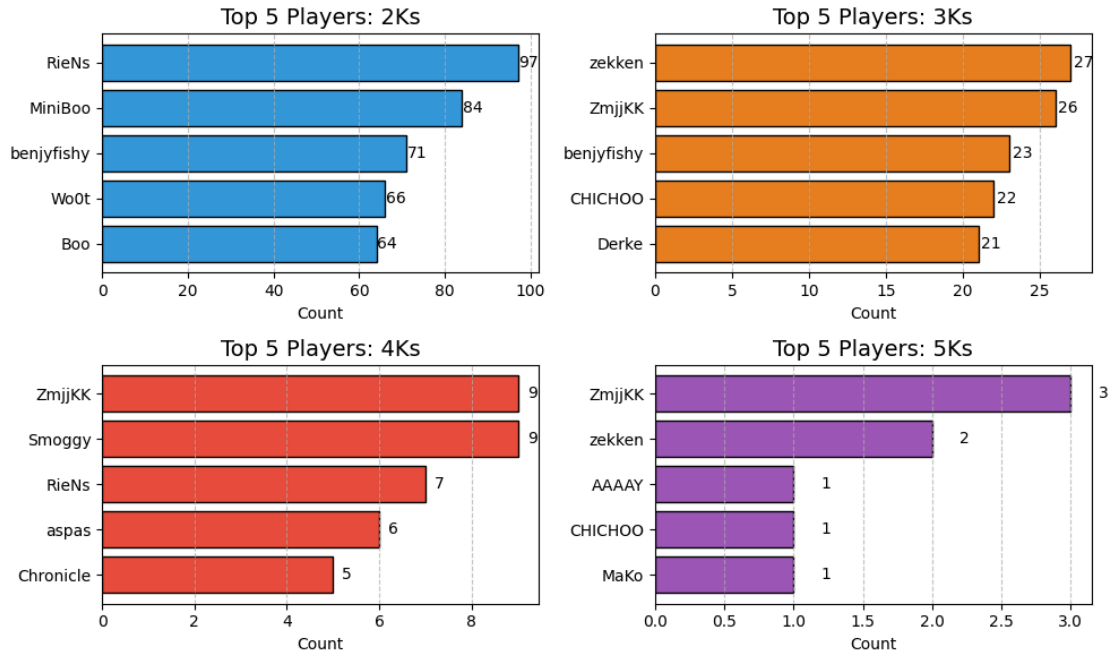


- Team Multi-Kill Distribution



Multi-kills follow the natural difficulty curve of competitive play — 2Ks (2267 rounds) and 3Ks (680 rounds) are common and form the backbone of impact, while 4Ks (158) and 5K Aces (14) are rare highlight moments requiring exceptional skill or opportunity. At the team level, PRX, FUT, and FNC rely heavily on consistent 3K rounds, whereas GEN, BLG, EDG, and DRX show higher shares of 4K/5K rounds, signaling reliance on explosive individual performances. This contrast highlights two strategic identities: teams built on steady, repeatable multi-kill impact versus those leaning on star players to deliver rare but game-changing heroics.

- Top 5 Players by Multi-Kill Type**



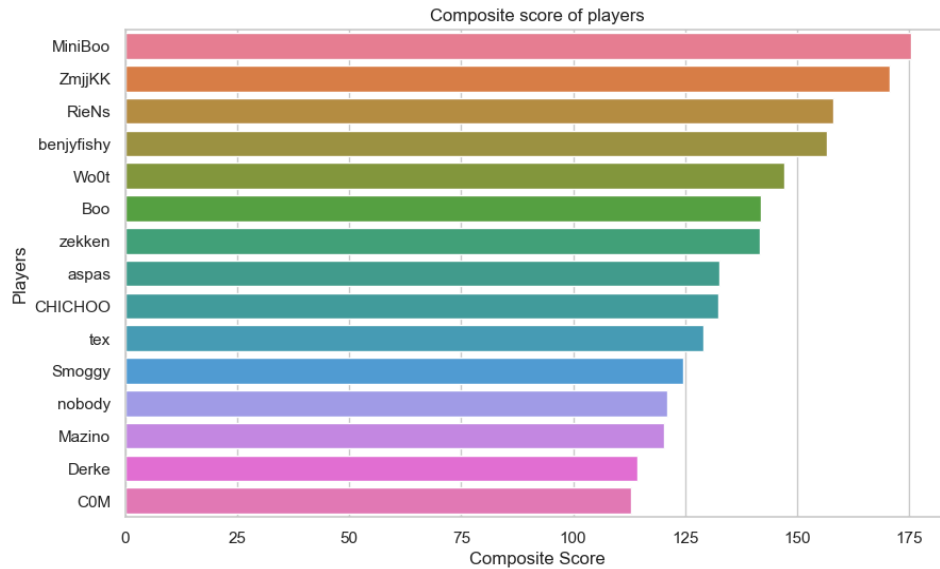
RieNs leads in 2Ks (97), showing consistency in smaller multi-kills, while zekken (27) and ZmjjKK (26) dominate 3Ks as frequent round-changers. For higher impact, ZmjjKK and Smoggy top 4Ks (9 each), and ZmjjKK again leads in 5Ks (3 aces), proving his explosive star power. Overall, players like RieNs and benjyfishy stabilize rounds with steady 2Ks/3Ks, while ZmjjKK and zekken deliver rare but decisive 4K/5K heroics.

● Top Players sorted by composite score

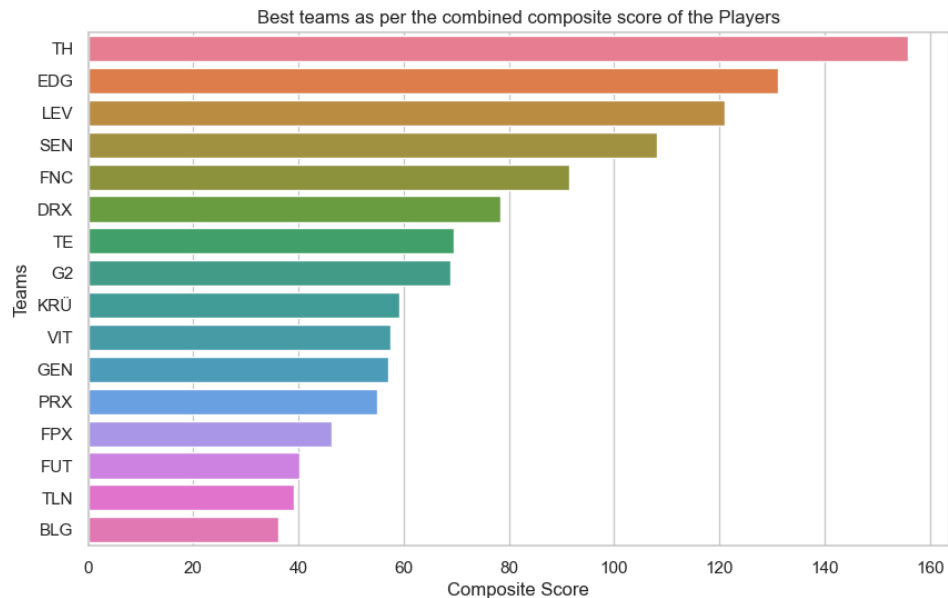
The composite score is a weighted metric designed to capture overall player impact by blending multiple performance factors. The weights are:

- First Kills (fk): 0.20
- First Deaths (fd): 0.20
- Kills: 0.15
- Deaths: 0.15
- Average Damage per Round (adr): 0.15
- Assists per Round (apr): 0.15

This balanced weighting ensures that both fragging power (kills, ADR, first kills) and survivability/utility (assists, minimizing deaths, avoiding first deaths) contribute to a player's overall score.



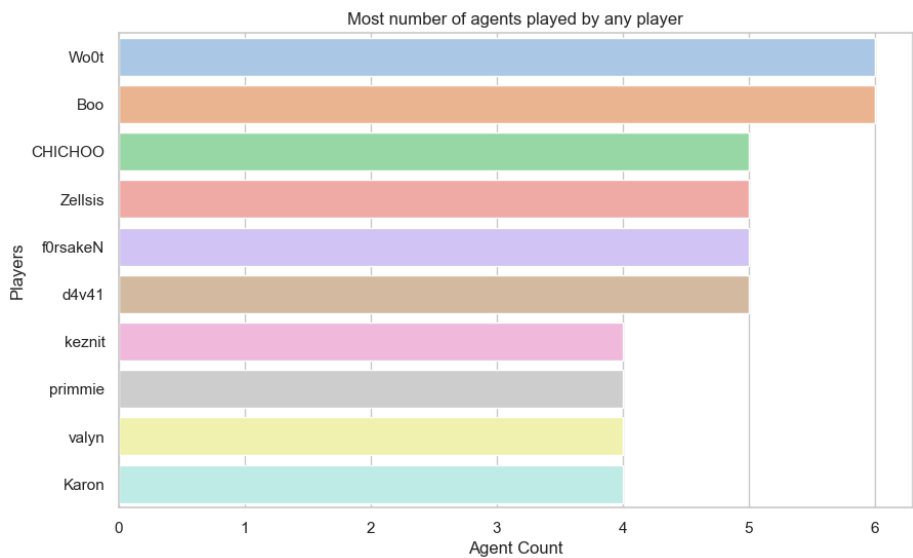
- Best teams as per the combined score of the players



The composite score blends weighted metrics — first kills and first deaths (20% each), kills, deaths, ADR, and assists per round (15% each) — to measure overall impact. At the player level, MiniBoo leads, with ZmijKK and RieNs close behind, while others like benjyfishy, Wo0t, and Boo also show strong balance; lower scorers such as Derke and C0M highlight how star fraggers can excel in specific areas but lack all-round consistency. When aggregated to the team level, TH (~155), EDG (~135), and LEV (~125) emerge as the most complete rosters, followed by SEN (~115) and FNC (~105), with DRX, TE, and G2 forming the mid-tier and BLG, TLN, FUT trailing at the bottom. These scores align closely with tournament placements: EDG finished 1st, TH 2nd, LEV 3rd, SEN 4th, DRX and FNC tied for 5th–6th, and TE and G2 tied for

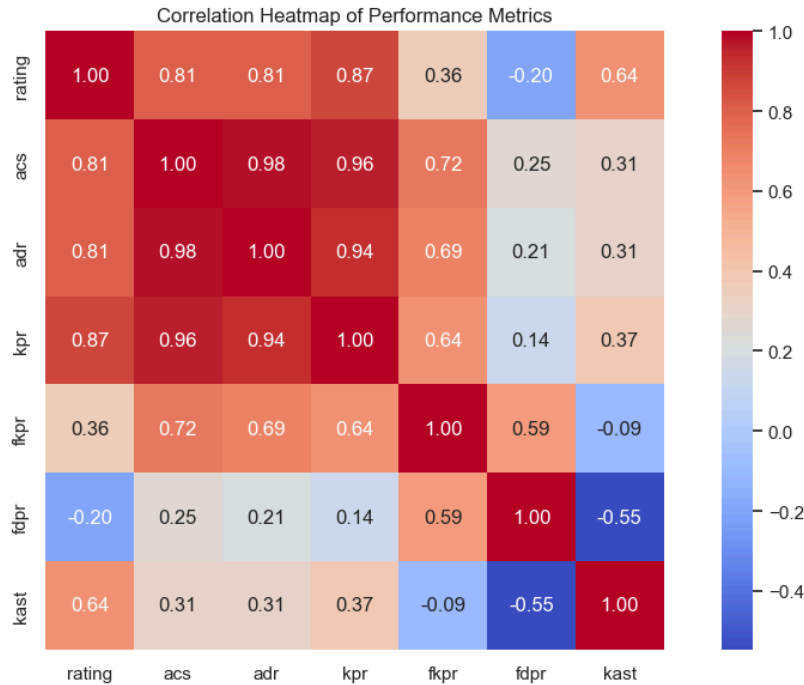
7th–8th, showing that higher composite depth strongly correlates with top finishes. The conclusion is clear — teams with balanced rosters and standout all-rounders (like MiniBoo and ZmjKK driving EDG and TH) consistently convert metrics into results, while squads with weaker composite depth struggle to translate isolated star power into sustained success.

- **Most agents played by any player**



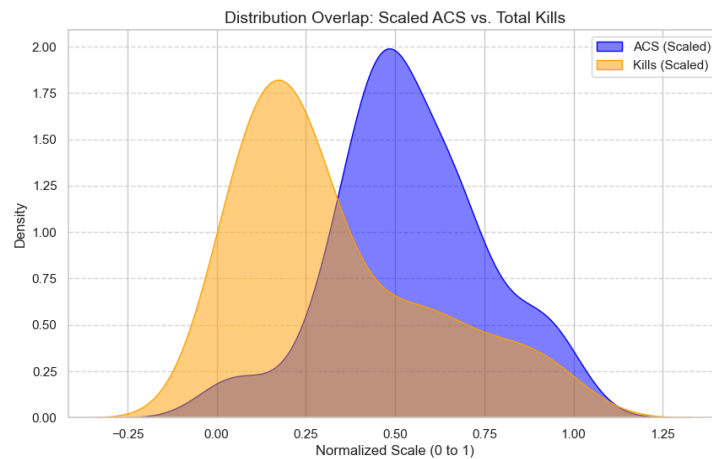
This chart highlights player versatility in agent selection, with Wo0t and Boo leading at 6 agents each, showing maximum flexibility and adaptability to team needs. Close behind, CHICHOO, Zellsis, f0rsakeN, and d4v41 each played 5 agents, reflecting strong versatility while still maintaining some specialization, while keznit, primmie, valyn, and Karon sit at 4 agents, showing moderate flexibility but leaning more toward defined roles. Overall, players with wider agent pools provide their teams greater tactical options across maps and opponents, while those with narrower pools excel in specialized roles but limit adaptability, meaning the most resilient teams are those that balance versatile flex players with role specialists.

- **Correlation Heatmap of Performance Metrics**



The heatmap shows that fragging and damage metrics (ACS, ADR, KPR) are the strongest drivers of performance rating, with very high positive correlations. First kills (FKPR) help but are less decisive, while first deaths (FDPR) and low KAST have clear negative impacts. In short: consistent kills and damage boost ratings, while early deaths and poor survivability drag them down.

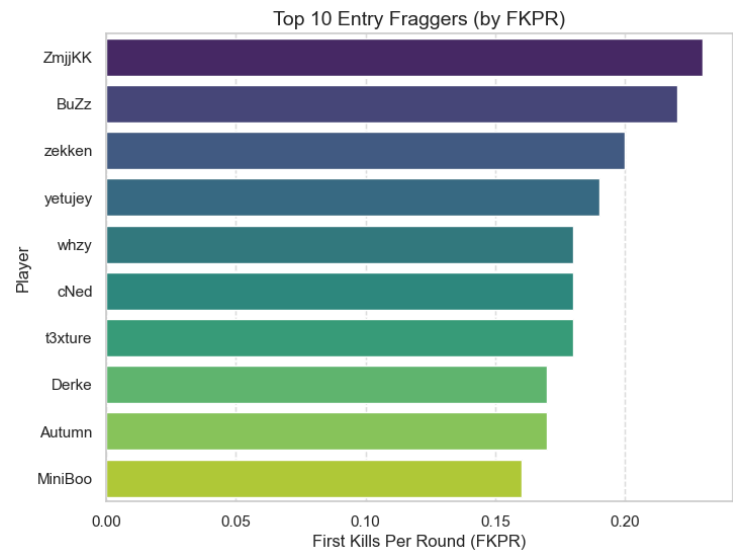
- **How Closely Do ACS and Kills Track?**



This density plot compares the distributions of scaled ACS (Average Combat Score) and total kills, both normalized between 0 and 1. The curves overlap significantly, showing that ACS and kills track closely together — players with higher kill counts generally also achieve higher ACS.

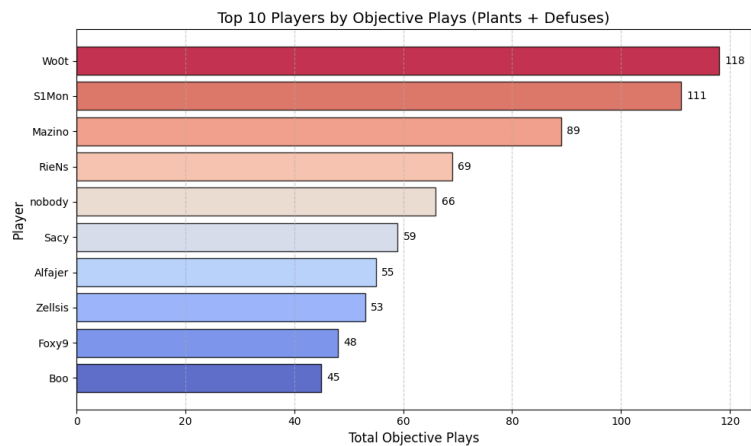
However, the slight differences in curve shape suggest ACS captures more than just raw kills, incorporating damage and utility contributions.

- **Top 10 Entry Fraggers**



The overlap between scaled ACS and total kills shows that fragging output strongly drives combat score, while the Top 10 entry fraggers chart highlights players like ZmijjKK, BuZz, and zekken as the most impactful openers who consistently secure first kills. Together, these insights confirm that while ACS reflects broader contributions like damage and utility, early-round fragging power from top entry players is a decisive factor in shaping both individual performance metrics and team momentum.

- **Top 10 Players by Game Objective Plays (Plants + Defuse)**



This chart shows the Top 10 players by objective plays (plants + defuses), with Wo0t (118) and S1Mon (111) leading by a wide margin, followed by Mazino (89), RieNs (69), and nobody (66) as consistent contributors, while Sacy, Alfajer, Zellsis, Foxy9, and Boo round out the list with

solid but lower totals. The conclusion is that objective plays are crucial for converting rounds into wins, and players like Wo0t and S1Mon stand out as the backbone of their teams' tactical execution, proving that success depends not only on kills but also on consistently delivering the plays that secure victory.

8.3 Key Findings

- **Consistent impact outweighs raw fragging volume.**
Players like Derke, zekken, and ZmjjKK lead in kills and ACS, but others such as trexx, Chronicle, and CHICHOO top ratings through efficiency, survivability, and utility impact. This shows that elite performance is driven by how reliably value is generated each round, not just by kill counts.
- **Damage output is the strongest driver of individual performance.**
ADR shows an almost perfect correlation with ACS, confirming that sustained damage across rounds is a more reliable indicator of impact than isolated kills. Players who consistently deal damage maintain higher combat scores even without leading frag charts.
- **Player roles introduce clear and predictable trade-offs.**
Aggressive players push both kills and deaths upward, reflecting high-risk, high-reward playstyles, while support players trade K/D for assists, objective impact, and utility value. Performance metrics must therefore be interpreted within role context rather than in isolation.
- **Winning maps amplifies individual efficiency across all roles.**
Average player ratings rise significantly on winning maps compared to losses, showing that consistent individual performance directly translates into team success. This pattern holds true for both fraggers and support players.
- **Clutch efficiency separates elite teams under pressure.**
Teams such as DRX, EDG, and FNC consistently lead in clutch percentage, demonstrating superior decision-making in high-pressure situations.
- **Multi-kill distributions reveal distinct team identities.**
Some teams rely on steady, repeatable 2K–3K contributions, while others depend on explosive 4K–5K hero rounds from star players. Teams built around consistent multi-kill output or having 1-2 players in a team that can consistently clutch and turn the tide of a match are required for greater stability and success across long series.
- **Balanced rosters outperform star-dependent lineups.**
Composite player scores align closely with final tournament standings, with teams like EDG, TH, and LEV succeeding due to depth and all-round contribution. Isolated star power alone is insufficient for sustained success.

- **Objective play is a critical but undervalued success factor.**
Players leading in plants and defuses consistently enable round conversions, reinforcing that winning Valorant depends as much on objective execution as on fragging power.
- **Player versatility increases strategic resilience.**
Teams with flexible players capable of operating across multiple agents gain greater adaptability in map vetoes and compositions. The strongest teams balance versatile flex players with role specialists rather than relying exclusively on either.

9. Pillar 3: Economy & Round Win Impact

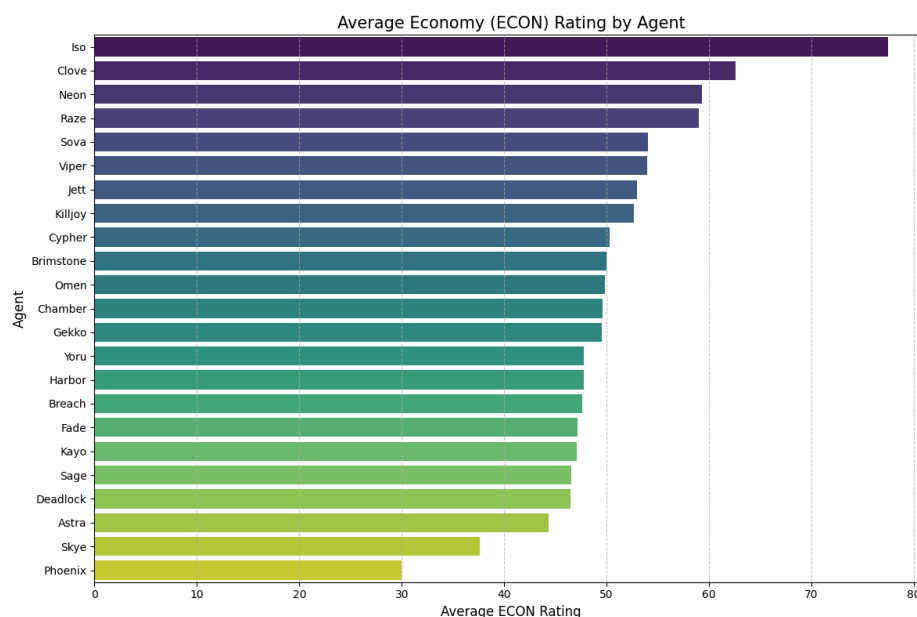
(Economy_stats.ipynb)

9.1 Analysis Focus

- Agent-level economic efficiency and cost effectiveness
- Round win probability across economy types (pistol, eco, semi-eco, semi-buy, full-buy)
- Impact of pistol and semi-buy rounds on map win probability
- Map-wise economy composition and spending patterns
- Team-level economic identities (full-buy reliance vs thrifty efficiency)
- Relationship between spending efficiency, momentum, and match outcomes

9.2 Analysis

- **Average ECON rating by Agent**

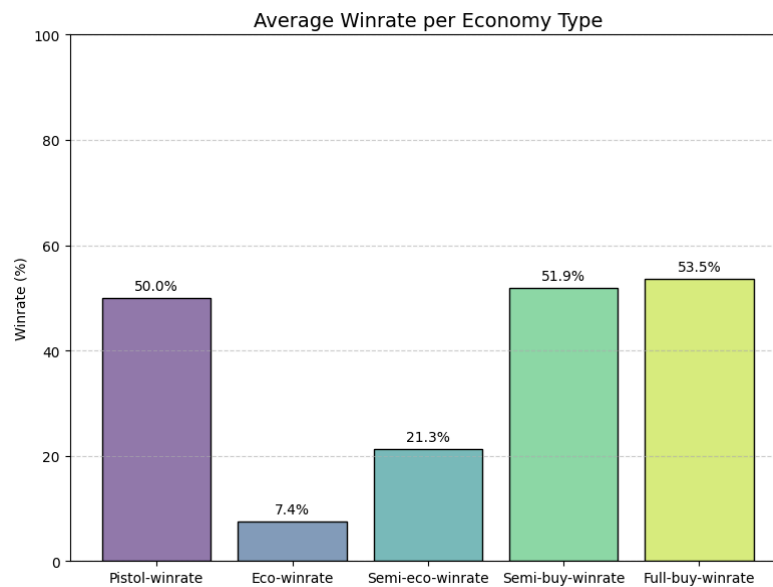


This chart shows the average ECON rating by agent, with Iso and Clove leading as the most economically efficient picks, followed by Neon and Raze, whose aggressive playstyles translate into strong cost effectiveness. Mid-tier agents like Sova, Viper, Jett, and Killjoy balance utility and fragging impact, while agents such as Phoenix, Skye, Astra, and Deadlock sit at the bottom, reflecting weaker economic returns. The conclusion is that agents with explosive fragging potential or versatile utility deliver the best efficiency, while more role-locked or situational agents lag behind, proving that agent choice directly shapes team economy and round win condition and long-term success.

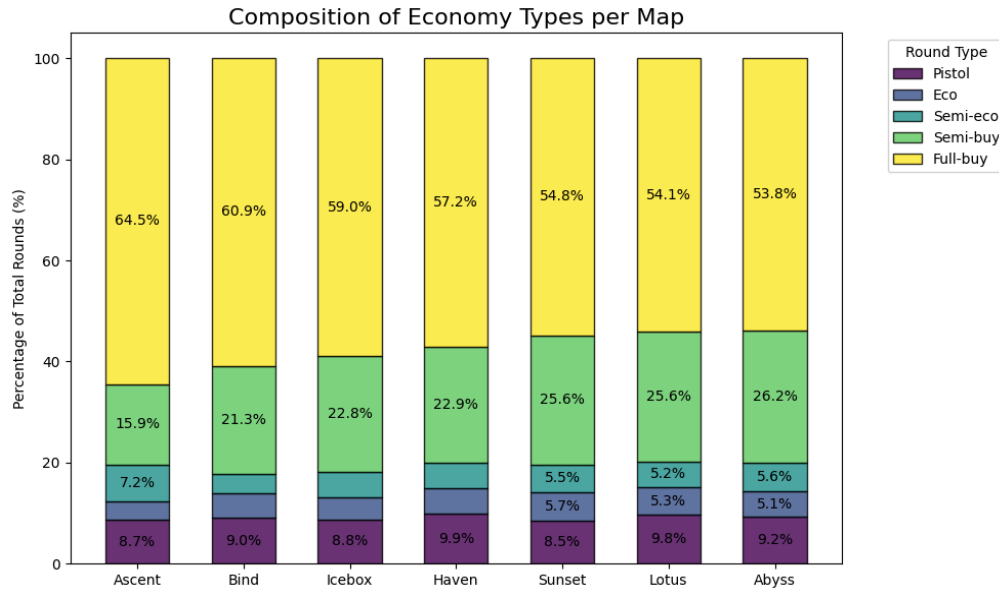
- **Average Winrate per Economy Type**

Economy Definations:

- Pistol: Early rounds with pistols only.
- Eco (0–5k): No significant spending, default pistols only.
- Semi-Eco (5–10k): Very light spending, saving for the next round.
- Semi-Buy (10–20k): Partial investment, mixing cheaper weapons with some utility.
- Full-Buy (20k+): Strongest setup with rifles, armor, and full utility.

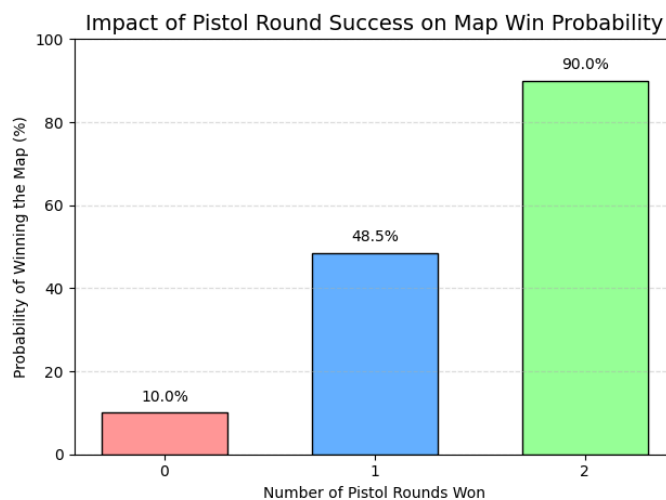


- **Composition of Economy Types per Map**

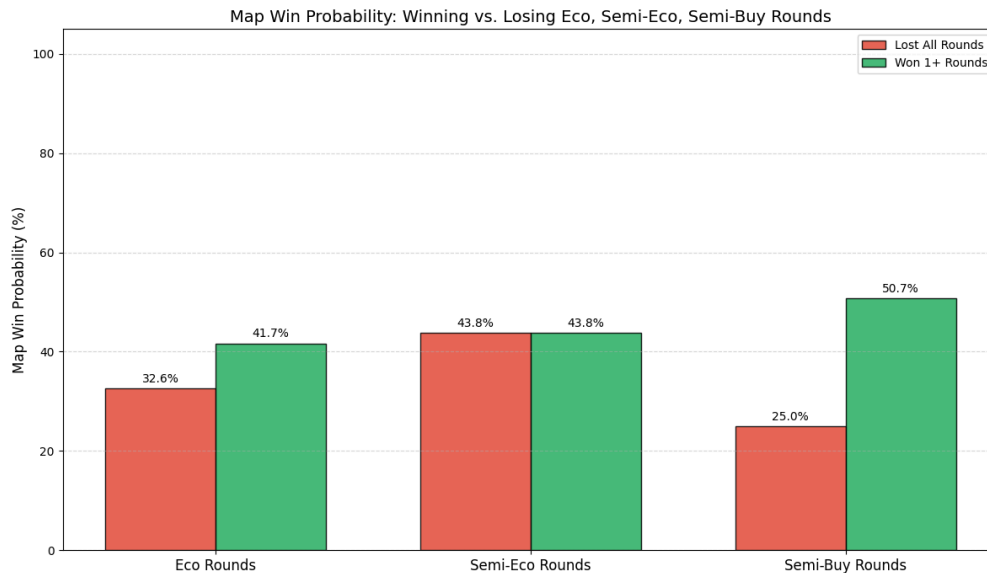


This chart compares average winrates across economy types, showing that full-buy (53.5%) and semi-buy (51.9%) rounds yield the highest success, while pistol rounds sit at 50%, and eco (7.4%) and semi-eco (21.3%) rounds trail far behind. When paired with the map-wise breakdown of economy composition, we see that full-buy rounds dominate across all maps (53–65%), while semi-buy rounds steadily increase from Ascent to Abyss, peaking at 26.2%, especially on newer maps like Sunset, Lotus, and Abyss. This trend suggests a strategic shift toward more frequent partial investments, where semi-buy rounds offer near full-buy win rates at lower cost. In contrast, eco and semi-eco rounds remain low-percentage setups, often used to reset economy or bait utility. Teams should prioritize full-buy setups but increasingly leverage semi-buy rounds as cost-effective win conditions, especially on maps with higher semi-buy frequency.

- **Impact of Winning Pistol Rounds on the Map Win Probability**

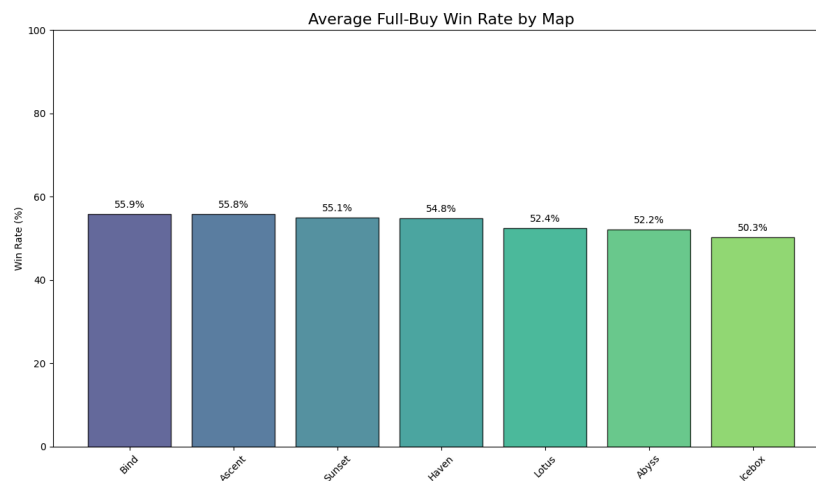


- **Match Win Probability : Winning vs Losing Eco, Semi- Eco, Semi- Buy Rounds**



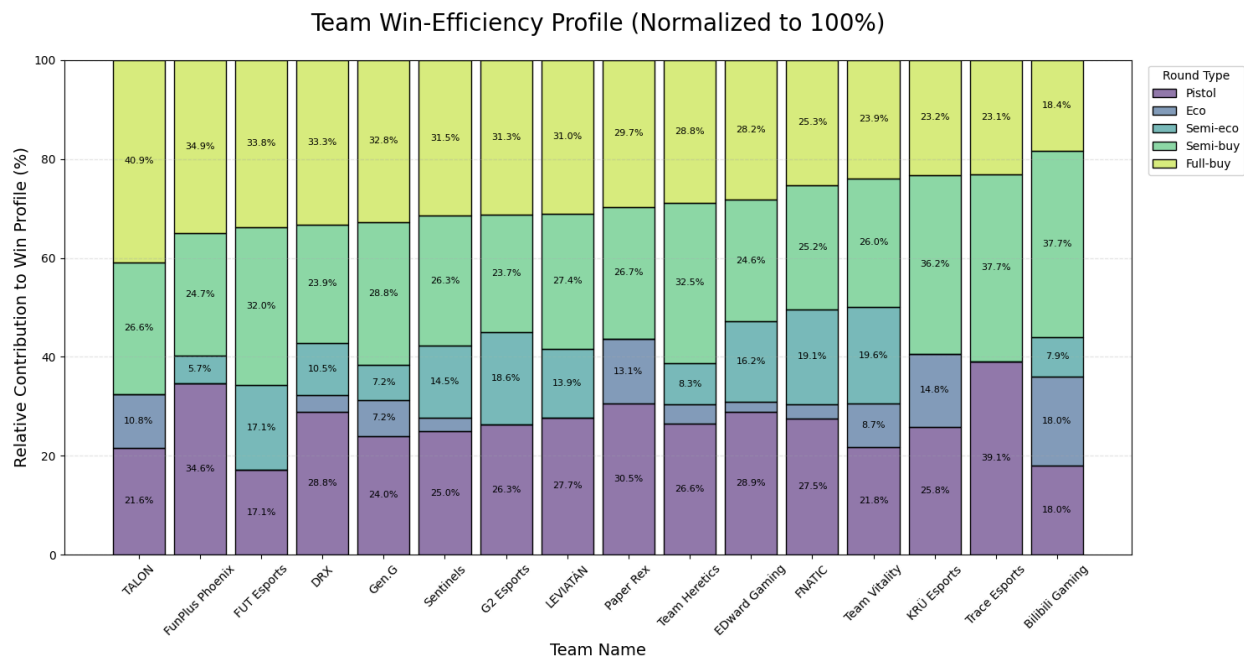
Winning pistol rounds and converting semi-buy rounds are easily the biggest swing factors in deciding map outcomes, with teams that fail to win any pistols left at just a 10% chance of taking the map, grabbing one pistol round pushing that up to 48.5%, and winning both skyrocketing it to 90%, which shows how much early momentum matters. Semi-buy rounds carry similar weight in the mid-game, where losing all drops map win probability to 25% but winning even one lifts it to 50.7%, almost on par with full-buy rounds. Ecos and semi-ecos don't look very impactful in the averages—32.6% vs. 41.7% for ecos, and 43.8% regardless of semi-eco outcomes—but the numbers don't really capture the moments where a single clutch eco or semi-eco win completely flips the map, breaking the opponent's economy, throwing off their rhythm, and adding pressure that can snowball into a win.

- **Average Full Buy Win Rate by Map**



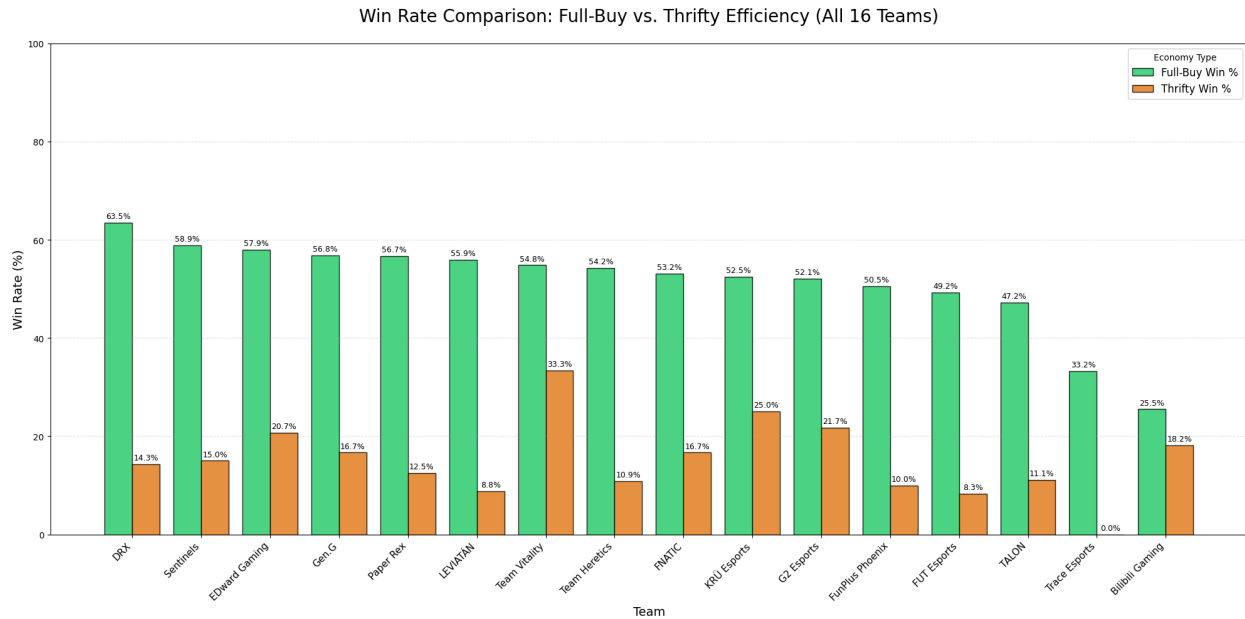
Looking only at full-buy win rates by map, we see a clear spread that highlights how map design and playstyle influence outcomes. Bind (55.9%) and Ascent (55.8%) sit at the top, showing that teams tend to convert their strongest setups more reliably on these maps, likely due to their balanced layouts and well-understood strategies. Sunset (55.1%) and Haven (54.8%) follow closely, suggesting strong consistency but slightly more variability in execution. On the lower end, Lotus (52.4%) and Abyss (52.2%) show reduced success, hinting at the complexity and newer meta challenges these maps present. Icebox (50.3%) sits at the bottom, reflecting how its verticality and unpredictable fights make even full-buy rounds harder to secure.

- **Team Winrate Profile by Economy Type**



Winning pistol rounds and converting semi-buy rounds are easily the biggest swing factors in deciding map outcomes, with teams that fail to win any pistols left at just a 10% chance of taking the map, grabbing one pistol round pushing that up to 48.5%, and winning both skyrocketing it to 90%, which shows how much early momentum matters. Semi-buy rounds carry similar weight in the mid-game, where losing all drops map win probability to 25% but winning even one lifts it to 50.7%, almost on par with full-buy rounds. Ecos and semi-ecos don't look very impactful in the averages—32.6% vs. 41.7% for ecos, and 43.8% regardless of semi-eco outcomes—but the numbers don't really capture the moments where a single clutch eco or semi-eco win completely flips the map, breaking the opponent's economy, throwing off their rhythm, and adding pressure that can snowball into a win.

- **Team Wise Full-Buy vs Thrifty Win Percentage**



Teams like DRX, Sentinels, and Edward Gaming excel with high full-buy win rates (above 57%) and rely heavily on structured, high-investment rounds, while their thrifty success remains modest (14–20%). Team Vitality stands out as an anomaly, converting only mid-tier full-buys (54.8%) but boasting a huge 33.3% thrifty win rate, with balanced contributions from semi-buy and eco rounds that highlight their strength in chaotic, low-econ scenarios. KRÜ Esports and G2 Esports also show solid full-buy conversion but punch above average in thrifty rounds (25.0% and 21.7%), supported by meaningful semi-buy impact. In contrast, Trace Esports and Bilibili Gaming struggle across both full-buy and thrifty rounds, leaning heavily on pistols or semi-buys and rarely recovering. Overall, the data reveals not just win rates but how teams construct their victories—some through consistency in full-buys, others through volatility and clutch potential in low-econ rounds.

9.3 Key Findings

- Higher spending increases win probability, but with diminishing returns.**
 Full-buy rounds deliver the highest win rates, yet their advantage over semi-buy rounds is relatively small. Once teams cross a minimum investment threshold, additional spending does not scale proportionally with success, particularly on maps with fast or chaotic engagements.
- Semi-buy rounds offer the strongest cost-to-value tradeoff.**
 Semi-buy rounds achieve near full-buy win rates at significantly lower cost and appear more frequently on newer and faster-paced maps such as Sunset, Lotus, and Abyss. This reflects a strategic shift toward flexible spending rather than rigid save-and-buy cycles.
- Pistol rounds are the largest momentum drivers in a map.**
 Winning pistol rounds dramatically increases map win probability, with teams converting

nearly 90% of maps when both pistols are won. Early-round economy control compounds quickly, making pistol preparation one of the most decisive factors in match outcomes.

- **Mid-game economy control outweighs eco outcomes.**
Winning at least one semi-buy round nearly doubles map win probability compared to losing all semi-buys, placing semi-buy conversion just below full-buy importance. Eco and semi-eco rounds, while impactful in isolated moments, primarily function as economy reset tools.
 - **Map structure directly shapes economic efficiency.**
Maps such as Bind and Ascent show higher full-buy conversion due to balanced layouts and predictable engagements, while Icebox, Lotus, and Abyss reduce full-buy reliability through verticality and faster rotations. Economy strategy must therefore adapt to map design.
 - **Teams win through distinct economic identities.**
Some teams rely on structured full-buy dominance, while others consistently extract value from thrifty and semi-buy rounds. Success is not defined by spending more, but by converting the right rounds at the right time.
 - **Efficient teams consistently win with lower average spend.**
Teams that balance disciplined full-buy play with strong semi-buy and occasional thrifty conversion outperform teams dependent on repeated high-investment rounds. Resource efficiency emerges as a clear competitive advantage.
-

10. Visualization Strategy

The visualization strategy prioritizes interpretability, fair comparison, and strategic relevance. Each chart type was chosen to directly answer a specific analytical question rather than to showcase volume or aesthetics.

Distributions were used to expose structural effects from tournament format, motivating the use of normalized metrics such as kills per round and ADR for unbiased player and team comparisons. Ranked bar charts enabled clear evaluation of agent efficiency, economy win rates, and team performance, quickly surfacing dominant patterns and outliers.

Boxplots and heatmaps captured variability and context dependency, highlighting how agent effectiveness and economy success shift by map rather than remaining uniform. These visuals revealed agent–map synergies, differences in pacing, and investment efficiency that averages alone would obscure.

Overall, the visuals function as decision-support tools, reinforcing the report's core insight: success at the highest level of Valorant is driven by efficiency, consistency, and informed strategic choices—not raw volume or isolated highlights.

11. Key Insights & Business Value

Strategic Insights

- Competitive success at Valorant Champions 2024 is driven by efficiency, consistency, and contextual decision-making rather than raw mechanical output alone.
- A small core agent pool defines the global meta, but optimal compositions are shaped by map structure and player fit, not pick-rate popularity.
- Teams that adapt agent choices to map-specific demands and individual strengths consistently outperform those relying on mirrored or static compositions.
- Player impact is best captured through normalized metrics such as ACS, ADR, and KAST, which reflect round influence, survivability, and contribution beyond kills.
- High-performing teams balance volatile, high-frag roles with stable, utility-driven players to ensure consistent value across rounds.
- Clutch potential and multi-kill efficiency matter most when supported by disciplined team structure rather than isolated hero plays.
- Economy management acts as a decisive competitive lever across maps.
- Winning pistol rounds and converting semi-buy rounds dramatically increases map win probability, often matching or exceeding the impact of full-buy rounds.
- Teams that consistently extract value from lower-investment rounds apply sustained economic pressure, forcing opponents into predictable buy cycles and reducing strategic flexibility.

Business Value

- Coaches can optimize agent compositions by map and assign roles based on efficiency metrics rather than reputation or raw frag counts.
- Analysts can evaluate player performance more accurately using normalized metrics, reducing bias caused by match volume or tournament longevity.
- Map veto and pick strategies can be reframed as risk-management decisions, helping teams minimize volatility across Bo3 formats.

- Strategic investment in semi-buy conversion and early-round planning delivers high returns with lower economic commitment.
 - The project demonstrates how structured data analysis converts complex in-game behavior into repeatable strategic advantages.
 - Performance efficiency, decision quality, and economic discipline emerge as scalable differentiators separating elite teams from the rest.
-

12. Technical Toolkit

Languages & Libraries

- Python as the core language for end-to-end analysis and modeling
- Pandas & NumPy for data cleaning, transformation, aggregation, and feature engineering
- Matplotlib & Seaborn for exploratory and explanatory visualizations
- Jupyter Notebook for iterative analysis, documentation, and reproducibility

Techniques

- Exploratory Data Analysis (EDA) to identify trends, outliers, and performance patterns
 - Metric normalization & aggregation to compare players and teams fairly across maps and match volumes
 - Correlation analysis to uncover relationships between performance, economy, and win outcomes
 - Composite scoring to quantify overall player and team impact using weighted metrics
 - Dimensionality reduction (PCA) to analyze player performance profiles and similarity
-

13. Limitations

- The analysis does not include round-by-round positional, gun, or utility usage data, limiting visibility into site control, trade patterns, utility efficiency, and how weapon choices influence round outcomes.

- Temporal dynamics such as momentum shifts, adaptation across halves, overtime behavior, and round-to-round strategy evolution were not explicitly modeled, which may underrepresent in-game adjustments.
 - Coach timeouts and tactical pauses are not captured in the dataset, preventing analysis of their immediate impact on round win probability, economy recovery, or momentum control.
 - External and qualitative factors—including team communication, pressure handling, crowd influence, and on-stage decision-making—are not measurable despite their significance in high-stakes matches.
-

14. Future Scope

- Add round-level gun and utility usage to evaluate loadout efficiency and ability impact.
- Incorporate positional data to analyze site control, rotations, and post-plant setups.
- Model coach timeouts and momentum shifts to quantify their effect on win probability.
- Isolate overtime and high-pressure rounds to study clutch performance under stress.
- Apply predictive models to estimate round and map outcomes based on economy and recent form.